

Science Fair Notebook

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Problem/Question

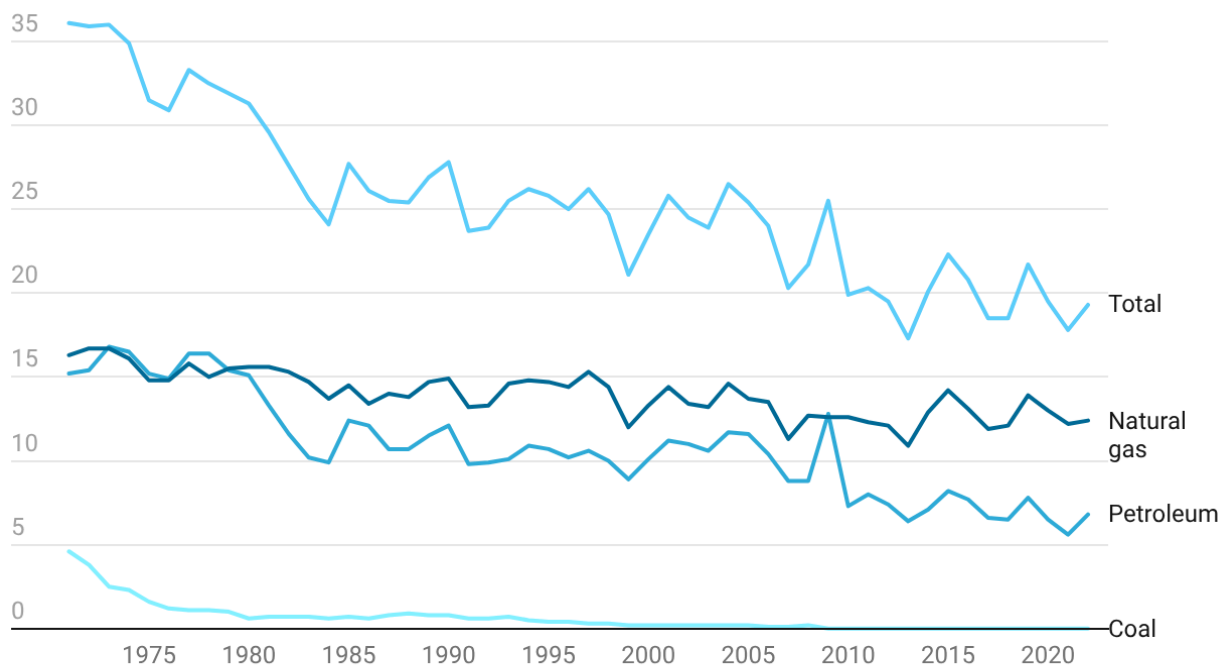
Which AI-MCDA architecture most accurately replicates physics-based ground truth for household energy decisions, while still maintaining reasonable failure rate and API costs?

Origin of Idea

I have wanted to make some kind of impact on worldwide emissions, and I was researching which sectors account for a large amount of emissions while also being fairly modifiable. I found that household energy choices contribute about 20% of carbon emissions in the US. However, most people make these choices using heuristics, which leads to non-optimal emissions while providing no further pleasure. I also found that asking people to change their habits had psychological barriers—specifically time use and ease of use. Thus, to be able to give customized recommendations for each house, I looked towards AI.

Research

The average U.S. household emits 8744 pounds of CO₂e at about 881 kWh per month (U.S. Environmental Protection Agency [EPA], 2025) and uses 10,791 Kilowatt Hours of energy (U.S. Energy Information Administration [EIA], 2022) annually. Including transportation, food, etc., per capita, the numbers change to about 17.9 tons of CO₂e (Center for Sustainable Systems, University of Michigan, 2025), and 81767 Kilowatt Hours of energy annually (EIA, 2022). The average U.S. household can save \$200-\$400 annually by optimizing their house to ENERGY STAR recommendations (DOE, n.d.). Figure 1 (EIA, 2022) visualizes the emissions of the residential sector.

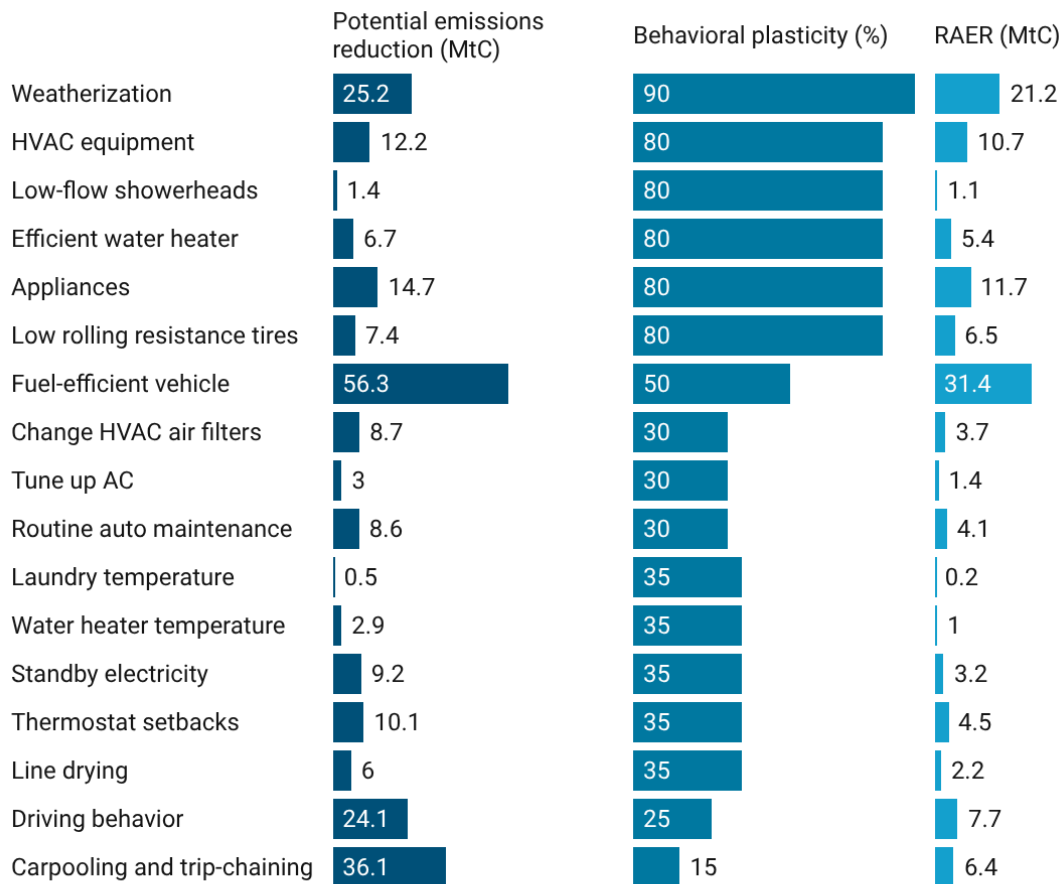


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While the average U.S. household consumes 10,791 kWh annually, this consumption is not evenly distributed. Space heating accounts for approximately 43% of residential energy use, followed by appliances (13%), water heating (12%),

and air conditioning (8%) (EIA, 2020). This distribution highlights key intervention points for emissions reduction. National reform of certain key household practices could result in a 123 million metric tonne (MtC) reduction in carbon per year as of 2005, when residential emissions accounted for 38% of overall US CO₂ emissions, or 626 MtC.

Additionally, these changes would result in little to no reduction in satisfaction, a necessary factor in widespread implementation. Dietz et al. (2009) demonstrated that household-level behavioral changes, if widely adopted, could cut U.S. emissions by 123 MtC annually. However, concerns for behavioral plasticity—how likely people are to implement each action based on financial and social incentives—remain an important factor. Figure 2 (Dietz et al., 2009) displays the behavioral plasticity, net carbon effect, and Reasonably Achievable Emission Reduction (RAER)



Created with Datawrapper

A major challenge with implementing an MCDA-type framework is the extra work required from users; although the benefits may be clear, some groups may not desire a switch in their routine, and have to wait for and consult a generated framework constantly. The decisions that the frameworks would be used for, such as purchasing a specific product or adjusting the thermostat, are often made using heuristics.

Heuristics—quick decision making—are useful in everyday life, but under complex or high-stakes situations, they can lead to errors; multiple studies have found that under time pressure, people default to “satisficing” (choosing the first

option that meets minimum criteria) or using dominant past behaviors, even if they're not ideal (Klapproth et al., 2008). Additionally, when these time constraints are combined with information overload, decision errors increase by approximately 20% more errors in judgement (Hahn et al., 1992). Since energy-use decisions often occur in high-distraction environments, tools that rely on sustained attention or complex input formats may fail to influence user behavior effectively. Very limited empirical research has been conducted thus far on using decision frameworks (especially MCDAs) in residential sectors.

AI is increasingly being used to create decision frameworks such as MCDAs. It has shown promise in the Marketing and Medical industries. (Jackson et al., 2024; Huang & Rust, 2020; Simon et al., 2024). Several studies have implemented frameworks, such as LCAs, MCDAs, MCDMs, Pareto Analysis, etc. successfully in environmental optimization decision making at a corporate or larger scale. AI's ability to manage large datasets suggests it could scale down to personal decision-making as well.

There is one substantial problem: constructing a framework typically requires extensive knowledge of MCDA construction methods (depending on the situation, certain ones often perform better than others), rubric design, and knowledge of the decision at hand. To remedy this, Huang and Burgherr (2024) created a calculator that assesses the situation and the criterion, and is able to evaluate the result. Although much more efficient than before, it still requires users

to understand how to diagnose a situation's problems and be able to format them appropriately. As a result, they remain inaccessible to the average household.

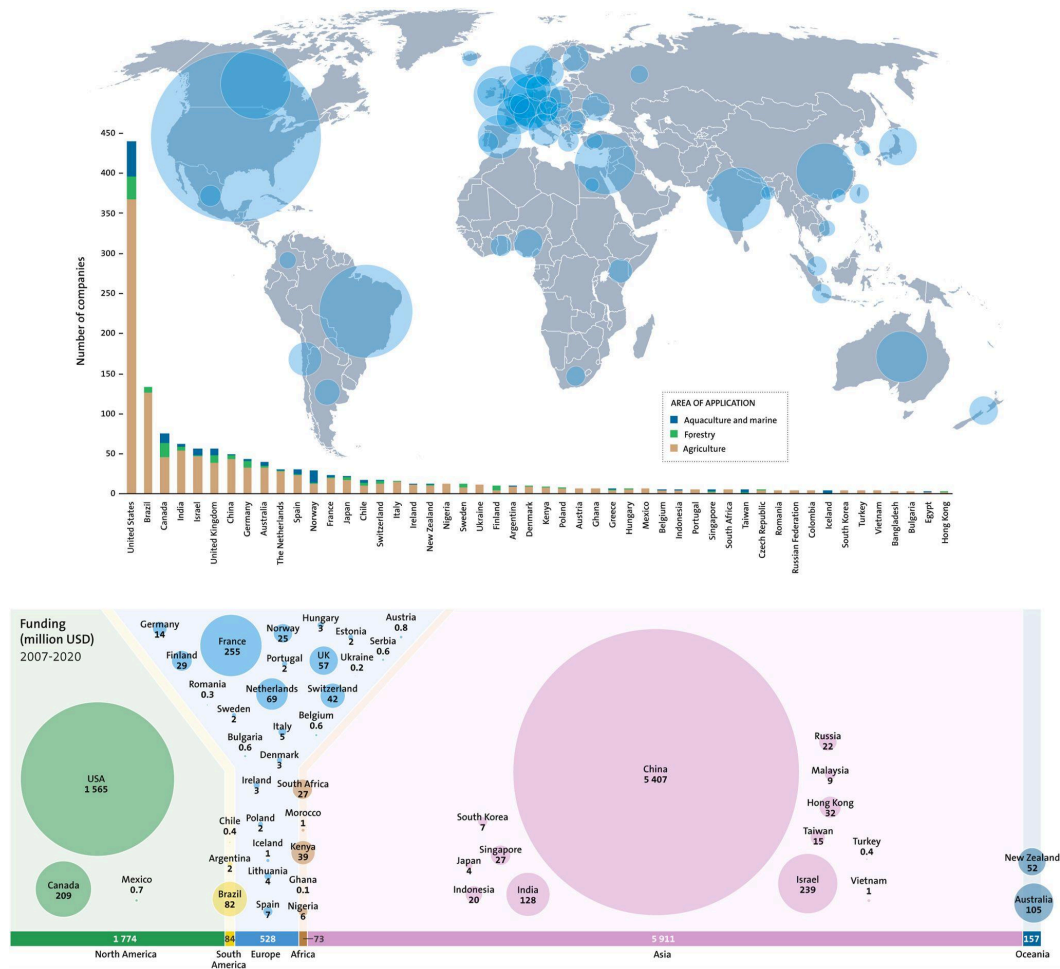
Using an AI in conjunction with an MCDA analysis tool would help remedy this issue; it would allow a participant to describe their situation in normal English, convert it to a series of alternatives with weights, taking into account the local weather conditions, electricity rates, etc. through APIs.

AI has been used to create MCDAs before by Igor Svoboda and Dmytro Lande (2024) in a study on cybersecurity analysis. They found that GPT-4 had a consistency ratio well below the accepted ratio of 0.1, which means that the decision outputs were coherent. They also tested a number of other AIs, such as GPT-3, which was found to be unreliable. This study tested and proved that AI models are reliable enough to use with decision frameworks, helping reinforce the plausibility of my research.

Modern systems also have a general knowledge of the energy use and carbon footprint of most products, among other metrics. AI has the ability to deduct these statistics from its own information, and also search the internet for useful information. However, AI has not been rigorously tested for individual-level decisions using longitudinal performance data.

Additionally, AI systems tend to create potential biases by assuming pieces of knowledge, such as assuming the same climate patterns or energy sources apply across all locations.(Galaz et al., 2021; Nishant et al., 2020). Figure 3 (Galaz et al.,

2021) illustrates how geographic concentration in training data can lead to skewed AI performance.



Thus, while AI systems should have a decent capability to understand contexts and predict scores based on knowledge, having a RAG database and/or a calculator to actually help back up their reasoning would likely help prevent multiple biases. This framework has not been tested in a real-world individual context, so empirical data is limited. Overall, AI shows significant promise for enhancing environmental decision-making at the individual level. This study aims to contribute early insights into how such tools can be structured and deployed effectively.

The convergence of these three research areas (household emissions, behavioral barriers, and AI-assisted decision frameworks) point to a clear opportunity. MAVT offers a mathematically sound way to compare tradeoffs between aspects of practicality, comfort, cost, and the environment. The natural language interface offered by LLMs eliminates the technical obstacle to utilizing these frameworks. The remaining questions are which integration architecture best maintains the mathematical rigor that initially makes MCDA valuable and whether LLMs can score criteria accurately enough to be trusted. Without addressing this, any tool based on these principles runs the risk of making confident but inaccurate recommendations, which could be worse than having no tool at all.

This study directly addresses that question by comparing three architectures with varying levels of assistance against calculations made with further specified information. This study offers the first empirical comparison of LLM integration in three different contexts—Pure Prompting, RAG-Enhanced, and Hybrid parameter extraction—specifically for household energy decision-making by testing across 100 scenarios, determining which method best replicates an actual expert’s MCDA scoring.

Hypothesis

The hybrid architecture (AI parameter extraction + calculator) will achieve lowest Mean Absolute Error (MAE) and Root Mean Square Error (RMSE), highest overall correlation in alternative ranking using Kendall's Tau and Spearman's rho with ground truth rankings, best match rate in alternative selections using Top-1 and Top-2, higher API token usage than the pure prompting architecture but lower usage than the RAG optimized architecture, and latency fast enough for household decision-making.

Variables

Independent Variable:

LLM Architecture type

Dependent Variable:

The accuracy of criterion scores, which was measured by MAE, RMSE, Kendall's tau, Spearman's rho, Top-1, and Top-2, and the efficiency, which was measured by tokens used, latency, and API cost per decision.

Constants:

- MAVT calculation method and the weights, which were Environmental Impact 35%, Energy Cost 30%, Comfort 20%, Practicality 15%
- MAVT value function types per criterion
 - Environmental impact: linear
 - Energy Cost: linear
 - Comfort: Logarithmic, $\alpha=1.5$
 - Practicality: Logarithmic, $\alpha=1.2$
- LLM model, which was Mistral Small 3.2 24B
 - Temperature set to 0.3, which along with time constraints did not allow 3 runs per architecture
- Number of RAG chunks retrieved per query (3)
 - Retrieved with ChromaDB
 - Using Sentence-Transformers model
- API calls per scenario per architecture; Pure and RAG had 3 calls each, and Hybrid had 1
- Reference ranges for normalization to 0-10 scale
- 100 total test scenarios (60 HVAC, 25 Appliance, 15 Shower)
- 50 separate scenarios for RAG database only (25 HVAC, 20 Appliance, 5 Shower)
- Pennsylvania energy parameters, which were \$0.19 per kWh electricity rate and an emissions factor of 0.6458 lbs CO₂ per kWh
- The time of use (TOU) pricing, which was calculated for off-peak at 60-80% of baseline and 180-270% for peak.
- Temperature comfort ranges, which were 73-79°F for cooling and 68-75°F for heating
- Reference Ranges and other important values available in Appendix

Experiment: Procedure

Materials:

Laptop, Python 3.11 programming environment, PyCharm IDE,

Dependencies: pandas 3.0.1, numpy 2.4.2, scikit-learn 1.8.0, scipy 1.17.1, sentence-transformers 5.2.3, transformers 5.2.0, torch 2.10.0, chromadb 1.5.2, openpyxl 3.1.5, and related dependencies (full list in requirements.txt).,

LLM API from Openrouter, EPA emissions data, EIA electricity consumption statistics, regional grid emissions data (full references for constants in bibliography)

Procedure:

1. Choose decision making types (I chose HVAC, Appliance, and Shower types)
2. Choose criterion for MCDA and choose weightings for each criterion. Choose LLM; I chose Mistral Small 3.2 24B through OpenRouter as the LLM for all three architectures, with a temperature setting of 0.3 to decrease variability.
3. Create a set of scenarios
 - a. Choose amounts for each decision type by priority
 - b. Choose variables needed for calculations, in order to get an accurate picture of the energy consumption, environmental impact, etc. For my three decision types, I needed
 - i. HVAC: Question, location, square footage, insulation, household size, utility budget, housing type, outdoor temperature, house age, R-value, HVAC age, SEER, occupancy context, alternative 1, alternative 2, and alternative 3.
 - ii. Appliance: Description, location, utility budget, appliance, housing type, occupants, peak rate, off-peak rate, kWh per cycle, appliance age/type, baseline time, alternative 1, alternative 2, and alternative 3.
 - iii. Shower: Description, location, occupants, tank size, gallons per minute (GPM), utility budget, housing type, outdoor temperature, water heater temperature, alternative 1, alternative 2, and alternative 3.

- c. Create scenarios by first choosing the variables for each of the scenarios, then constructing an overall question based off of the variables chosen
- d. Create a set of test scenarios by scaling down/generalizing information provided so that the AI has to make reasonable assumptions
 - i. For my three decision types, I removed from GT list (other things were generalized to ranges/words:
 - 1. HVAC: R-value, HVAC age, SEER, and occupancy context
 - 2. Appliance: appliance, peak rate, off-peak rate, kilowatt-hours per cycle, appliance age, baseline time
 - 3. Shower: tank size and water heater temperature
 - ii. Make sure to leave scenarios for a future RAG database (I chose to keep 50 extra, which are not used in the test set. This leads to a total of 150 scenarios)
- 4. Create Ground-truth calculators for each decision type
 - a. Make functions to calculate the scores of each of your criterion
 - i. Make sure to first research reference ranges; what are the 5th and 95th percentile of values for the criterion you are calculating (e.g. carbon emissions & electricity usage for a dishwasher)
 - ii. Then, research formulas; for behavioral criterion, understand how people react to certain times/usages/etc. For criterion with actual values, such as energy cost, use your predefined variables to calculate kWh.
 - b. Create a MAVT function in each calculator (same for each). For value function type for each criterion, research how each criterion affects/is perceived by humans. For quantitative criteria, linear functions are appropriate. For behavioral criteria, a logarithmic function is better. My final selections were linear for energy cost and environmental impact, and logarithmic ($\alpha=1.5$ & 1.2 respectively) for comfort and practicality.
 - i. MAVT normalization will be done to each criteria based on your references ranges to a 0–10 scale by decision type
 - ii. Aggregate all 4 criterion values based on weights into a single weighted sum, and then have each of the 3 alternatives in each scenario be ranked by value of the weighted sum
- 5. Run ground truth calculators on all 150 scenarios
- 6. Implement three architectures. Use identical MAVT weights for all:
 - a. Pure Prompting: For each scenario, make 3 separate API calls (one per alternative). Each call provides the LLM only with the scenario text and asks it to output scores for all four criterion. No additional

context, formulas, or reference data is provided. Output scores are weighted and summed for ranking

- b. RAG-Enhanced: Same 3-call-per-scenario structure as Pure Prompting, but before scoring each alternative, retrieve the 3 most relevant chunks from a ChromaDB vector database containing the set aside 50 scenarios (mine were split into 25 HVAC, 20 Appliance, 5 Shower), which contain ground truth scores. Append retrieved context to the prompt before asking the LLM to score criteria. Output scores are weighted and summed for ranking
 - i. Create a program to build the RAG database by processing your 50 scenarios with ChromaDB for retrieval using the sentence-transformers/all-MiniLM-L6-v2 alternative.
 - c. Hybrid: Make 1 API call per scenario (not per alternative). The LLM is asked only to extract parameters from the scenario text (such as setpoint temperature, flow rate, appliance type, usage time, etc.). Those parameters are passed directly to the LLM's selected ground truth calculator from step 4, which computes all criterion scores based on the given parameters. MAVT weighting is then applied identically to ground truth calculation method.
7. For each architecture, compare output rankings against ground truth using Kendall's tau, Spearman's rho, Top-1 accuracy, Top-2 accuracy, MAE, and RMSE per criterion. Record total tokens, latency, and cost per decision for efficiency comparison.

Experiment: Data

Day + Time worked	What was accomplished
11/20/25 2 hours	I started looking into project ideas. I talked to one of my friends who has researched before to refine my idea. I started with wanting to do some kind of environmental engineering project. As of right now, I have narrowed it down to a coding project, since he told me that getting some kind of laboratory research on such a short notice was infeasible. I also looked into doing a project in watershed testing.
11/23/25 3 hours	I started today out by looking into data science projects related to environmental conservation. I found <i>Multi-criteria decision analysis in environmental sciences: Ten years of applications and trends</i> , which looks like an interesting direction to go in. The paper speaks on other commercial uses, but what I have noticed from this source and other sources is that there isn't much mention of using MCDAs on a smaller scale, though I do see the barrier of actually creating an MCDA for a smaller group. Still, this idea seems interesting. I also found numerous factsheets from the government and universities on current emissions by sector
12/1/25 1.5 hours	I went back through my sources and noticed from the factsheets that households are actually a pretty large portion of emissions. I also found a source by a group of researchers led by Dietz that shows which household decisions in particular have high potential for emissions reduction.
12/2/25 45 minutes	I discussed ideas further with my friend and also asked chatGPT for some further inspiration on project ideas with MCDAs, specifically in households. I believe using some kind of AI to get over the logistical barrier could be feasible
12/4/25 5 minutes	I'm not actually currently at my laptop right now but I had an idea and just wanted to write it down so I don't forget it: what if I actually test the capability of AI in MCDAs itself?
12/10/25 4 hours	Today I did quite a few things. I am now thinking of having an AI look at scenarios, then come up with three different ways to go about the decision, then output scores for each of the criterion based on the question. I have also decided on evaluating this AI using a calculator. This calculator will actually have formulas and such code to be able to come up

	with a true score for each of the criterion on each of the decision paths. The scores the AI outputs will then be modified using the TOPSIS method of MCDA using scikit-criteria to actually come up with the best decision for that scenario (according to the AI) I could then compare the calculator's and the AI's choices to find out how accurate the AI was. The calculator would most likely need quite a few parameters, which may need to be generated by the AI I also emailed numerous professors for help with the project
12/11/25 45 minutes	First off, I looked into how our school registers for the CRSCC, and then emailed Mrs. King for more information on what she needs. I also realized that having the AI generate parameters leads to a flaw in the overall experiment; we are no longer evaluating whether the AI can score well, but now also how well it creates parameters, which is not my intent. I also heard back from Dr. River (he) Huang. We are scheduling a meeting for the future.
12/18/25 2 hours	Scheduled meeting with Dr. Huang on January 20th. I also continued to look into further sources. I have now started exploring sources on behavioral barriers to using MCDAs in household situations, which include time barriers, habit barriers, adoption barriers, etc. This has helped me reinforce my belief on evaluating how well AI can create MCDAs
12/20/25 2 hours	During my meeting with Dr. Huang, I modified my project heavily and he helped me understand what parts on my project were unfeasible/not testable, and how to change it so I could reduce the amount of independent variables. I also learned more about what kind of functions to use, and he recommended I switch to MAVT functions at the end due to the way I was planning on computing ground truth (using hard values to calculate things like energy cost in kWh, Environmental in lbs CO₂, and comfort and practicality according to times and noise barriers rather than arbitrary ratings). We changed these things: 1. Overall, while testing AI may help determine the feasibility of AI itself, by testing it against other versions (He suggested a RAG database for another architecture, and then using the AI in conjunction with the calculator in order to see its accuracy in parameter extraction) I would get a fuller range of where AI succeeds vs. where it needs some help 2. A ground truth calculator would need a large amount of parameters to actually give relevant scores. Not giving the AI any of the same parameters wouldn't be fair and would subject the experiment to

	failure on the question rather than the AI. Thus, I should also give the AI a reduced set of parameters (and/or generalized). 3. We decided on HVAC, Appliance, and Shower Decision types 4. He recommended at least 100 quality scenarios, and another 50 for the RAG database 5. Have 3 predetermined alternatives so that we are just testing AI'sd scoring
12/23/25 1 hour	Researched and comprehended MAVT as well as why it is useful. I also set up the project github and linked it to pycharm, created 4 base files, and started working on hardcoding MAVT. I also decided to use temperatures for the HVAC alternatives, times of use for the appliance alternatives, and water heat temperatures for the shower alternative
12/24/25 2 hours	Created MAVT library and the 4 value function types (linear, concave, convex, and piecewise). Also created limits on alpha, beta, and slope values to avoid extreme curves. I also started to add the basic functions of the pure-prompting file, such as the LLM pipeline and value functions request (the import was successful). I also realized that water heater temperatures for the shower alternatives doesn't actually line up with the decision type we are looking for (daily decisions) so I have modified it to shower length.
12/25/25 1.5 hours	I also started researching what value functions are needed for every decision type, and started the code for the function that decides what the function type should be. Chose Deepseek Exactus Termino. Added basic prompting for the pure-prompting AI file. I have also added in Openweather and OpenEI APIs for real time calculations on electricity rates and outdoor temperature.
1/1/26 2 hours	Finalized scenario information both for architectures and ground-truth calculator. Started creating all 100 scenarios, now divided into "simple" (only one variable changing) and "complex" (2+ variables change)\ EXAMPLE SCENARIO FOR AI:

```

{
  "scenario_id": "1",
  "problem": "What temperature should I set my AC to while I go on a 2 week vacation?",
  "Location": "Chester Springs, PA",
  "square_footage": 1800,
  "insulation": "poor",
  "household_size": 3,
  "utility_budget_per_month": 200,
  "home_type": "House",
  "age_of_home": "20+ years"
}

```

1/2/26
2 hours

Finished scenario corpus for GT and revised alpha + beta values for ground truth. Still have to remove scenario types that I will not be focusing on-I have to decide whether purchase/changing unit decisions will be included at all for a “complex” decision type. Outside of metrics, I will report token use, latency, and API cost

Question	Location	Square Footage	Insulation	Household Size	Utility Budget	Housing Type	Outdoor Temp	House Age	R-Value	HVAC Age	SI
What temperature should I set my AC to while I go on a 2 week vacation?	Chester Springs, PA	1800	Poor	3	\$200	Condo	88°F	20+ years	11	18	
What temperature should I set my AC to tonight?	Chester Springs, PA	1800	Poor	3	\$200	Condo	25°F	20+ years	11	18	
What temperature should I set my AC to tonight?	Lancaster, PA	2500	Good	4	\$500	Single-family	92°F	25+ years	19	8	
What temperature should I set my AC to tonight?	Lancaster, PA	2500	Good	4	\$500	Single-family	20°F	25+ years	19	8	
What temperature should I set my AC to tonight?	Pittsburgh, PA	2000	Medium	2	\$250	Single-family	85°F	30+ years	13	15	
What temperature should I set my AC to tonight?	Pittsburgh, PA	2000	Medium	2	\$250	Single-family	28°F	30+ years	13	15	
What temperature should I set my AC to tonight?	West Chester, PA	1600	Medium	2	\$220	Townhouse	90°F	35+ years	13	22	
What temperature should I set my AC to tonight?	West Chester, PA	1600	Medium	2	\$220	Townhouse	22°F	35+ years	13	22	
What temperature should I set my AC to tonight?	State College, PA	2100	Good	3	\$300	Single-family	86°F	40+ years	19	10	
What temperature should I set my AC to tonight?	State College, PA	2100	Good	3	\$300	Single-family	18°F	40+ years	19	10	
What temperature should I set my AC to tonight?	Philadelphia, PA	900	Poor	1	\$150	Apartment	93°F	45+ years	11	35	
What temperature should I set my AC to tonight?	Philadelphia, PA	900	Poor	1	\$150	Apartment	24°F	45+ years	11	35	
What temperature should I set my AC to tonight?	Erie, PA	1800	Medium	4	\$260	Single-family	82°F	20-24 years	13	12	
What temperature should I set my AC to tonight?	Erie, PA	1800	Medium	4	\$260	Single-family	15°F	20-24 years	13	12	
What temperature should I set my AC to tonight?	Bethlehem, PA	1400	Good	2	\$200	Rowhouse	89°F	22-24 years	19	6	
What temperature should I set my AC to tonight?	Bethlehem, PA	1400	Good	2	\$200	Rowhouse	26°F	22-24 years	19	6	
What temperature should I set my AC to tonight?	Doylestown, PA	2300	Good	4	\$350	Single-family	91°F	15+ years	19	7	
What temperature should I set my AC to tonight?	Doylestown, PA	2300	Good	4	\$350	Single-family	19°F	15+ years	19	7	
What temperature should I set my AC to tonight?	Media, PA	1700	Medium	3	\$240	Twin	87°F	10+ years	13	18	
What temperature should I set my AC to tonight?	Media, PA	1700	Medium	3	\$240	Twin	23°F	10+ years	13	18	
What temperature should I set my AC to tonight?	Allentown, PA	1900	Medium	3	\$280	Single-family	90°F	5 years	13	14	
What temperature should I set my AC to tonight?	Allentown, PA	1900	Medium	3	\$280	Single-family	21°F	5 years	13	14	
What temperature should I set my AC to tonight?	Harrisburg, PA	2600	Poor	5	\$320	Single-family	94°F	0-1 years	11	28	
What temperature should I set my AC to tonight?	Harrisburg, PA	2600	Poor	5	\$320	Single-family	17°F	0-1 years	11	28	
What temperature should I set my AC to tonight?	Blue Bell, PA	2800	Good	4	\$450	Single-family	88°F	0-2 years	19	5	
What temperature should I set my AC to tonight?	Blue Bell, PA	2800	Good	4	\$450	Single-family	22°F	0-2 years	19	5	

1/3/26
1 hour

Spent the entirety of today finishing the AI version of the scenario lists for each decision type.

1	Question	Location	Square Footage	Insulation	Household Size	Utility Budget	Housing Type	House Age	Complexity				
2	What temperature should I set my AC to while I go on vacation for 2 weeks?	Chester Springs, PA	1800	Poor	3	200	Condo	20+ years	Simple				
3	What temperature should I set my AC to tonight?	Lancaster, PA	2500	Good	4	500	Single-family	5 years	Simple				
4	Should I reduce my AC setpoint at night?	Pittsburgh, PA	2000	Medium	2	250	Single-family	10+ years	Complex				
5	Should I turn my AC off or raise the setpoint while I'm at work for 9 hours?	West Chester, PA	1600	Medium	2	220	Townhouse	15 years	Complex				
6	What temperature should I set my thermostat during a weekend heat wave?	State College, PA	2100	Good	3	300	Single-family	8 years	Simple				
7	Is it worth pre-cooling my house before peak electric rates?	Philadelphia, PA	900	Poor	1	150	Apartment	40+ years	Complex				
8	Should I run the HVAC fan continuously today?	Erie, PA	1800	Medium	4	260	Single-family	12 years	Complex				
9	What AC setpoint balances comfort and cost for sleeping?	Bethlehem, PA	1400	Good	2	200	Rowhouse	20 years	Complex				
10	Should I open windows instead of using AC this evening?	Doylestown, PA	2300	Good	4	350	Single-family	6 years	Complex				
11	How much should I raise my thermostat when leaving for a 3-day trip?	Media, PA	1700	Medium	3	240	Twin	18 years	Simple				
12	Is it okay to set my AC lower to cool faster after work?	Allentown, PA	1900	Medium	3	280	Single-family	14 years	Complex				
13	Should I close vents in unused rooms during the day?	Harrisburg, PA	2600	Poor	5	320	Single-family	25 years	Complex				
14	What's the best thermostat setting for a work-from-home day?	Blue Bell, PA	2800	Good	4	450	Single-family	4 years	Simple				
15	Should I adjust my thermostat if it's very humid but not hot?	York, PA	1500	Poor	2	180	Single-family	30+ years	Complex				
16	Is a small temperature setback worth it for just 2 hours?	Phoenixville, PA	2000	Medium	3	260	Single-family	10 years	Complex				
17	What AC temperature minimizes cost during an overnight low?	Scranton, PA	1750	Medium	2	210	Single-family	16 years	Complex				
18	Should I keep my AC running during a short afternoon outing?	Norristown, PA	1200	Poor	3	170	Apartment	35 years	Simple				
19	Is it better to cool the house early or wait until evening?	Lower Merion, PA	3000	Good	4	600	Single-family	7 years	Complex				
20	What thermostat setting is best when guests are over?	King of Prussia, PA	2200	Medium	5	400	Townhouse	9 years	Simple				
21	Should I use ceiling fans instead of lowering the AC?	Pottstown, PA	1850	Medium	3	230	Single-family	20 years	Complex				

1/4/26

Started work on HVAC ground-truth calculator. Switched AI to Mistral

1 hour	small 3.2 for better cost to reasoning ratio, and also read some documentation for Openweather, OpenEI, and Openrouter API keys
1/6/26 20 minutes	Added Scenarios to Pycharm from google sheets version and committed to Git. I also worked on refining the parameters for the HVAC scenarios; I added SEER and R values to the calculator, and then changed the overall insulation for the AI versions to just be “good/average/poor”
1/7/26 30 minutes	I was running the scenarios and kept getting errors, so I am currently working on diagnosing the error with the OpenEI key in retrieving electricity costs (it asks for Zip Codes but I tested a few different ones and it was returning the same cost) and I also updated the code to read in the HVAC age, which it wasn't taking in earlier giving another error
1/10/26 2 hours	Worked further on the ground-truth; some practicality and comfort scores were 0, and many of the other values were in a very strange range (<1 or >10) which I believe is due to normalizing before the value function is done being applied. I am working on fixing this right now, and am testing with certain HVAC scenarios currently. I hope to finish all 3 architectures and this ground-truth (without iterations or analysis) within 2-3 weeks; I believe I have till early february to submit project.
1/16/26 1 hour	Fixed MAVT implementation; I was working with overly complex functions. Dr. He Huang sent me a more tailored list to use:

Possible types of single-attribute value functions		
Type	Increasing value functions	Decreasing value functions
Linear	$u(x) = \frac{x - x_{min}}{x_{max} - x_{min}}$	$u(x) = \frac{x_{max} - x}{x_{max} - x_{min}}$
Polynomial	$u(x) = \left(\frac{x - x_{min}}{x_{max} - x_{min}} \right)^a$ For any $a > 0$: if $a > 1$, the function is convex downward (concave upward); if $0 < a < 1$, the function is convex upward (concave downward); a symmetric reflection occurs if $a \rightarrow \frac{1}{a}$.	$u(x) = \left(\frac{x_{max} - x}{x_{max} - x_{min}} \right)^a$
Exponential	$u(x) = \frac{1 - \exp\left(a \cdot \frac{x - x_{min}}{x_{max} - x_{min}}\right)}{1 - \exp(a)}$ For any $a \neq 0$: if $a > 0$, the function is convex downward (concave upward); if $a < 0$, the function is convex upward (concave downward); a symmetric reflection occurs if $a \rightarrow -a$.	$u(x) = \frac{1 - \exp\left(a \cdot \frac{x_{max} - x}{x_{max} - x_{min}}\right)}{1 - \exp(a)}$
Logarithmic	$u(x) = \frac{\ln\left(a \cdot \frac{x - x_{min}}{x_{max} - x_{min}} + 1\right)}{\ln(a+1)}$ For any $a > -1$: if $a > 0$, the function is convex upward (concave downward); if $-1 < a < 0$, the function is convex downward (concave upward); a symmetric reflection occurs if $a \rightarrow (1 - a)$.	$u(x) = \frac{\ln\left(a \cdot \frac{x_{max} - x}{x_{max} - x_{min}} + 1\right)}{\ln(a+1)}$
I also tested my first few scenarios successfully.		
1/24/26 3 hours	Worked exclusively on the HVAC ground truth; my thought process is that if I can get one ground truth calculator and 1 actual architecture done, the other 2 for each can be modeled off easily. Today I updated multiple references ranges (many of the old ones were placeholders) using 5th and 95th percentile for energy cost and environmental impact, and then focusing on ASHARE standards & other researched thresholds for comfort and practicality. I also realized that using OpenEI for electricity rates has no real effect on the scores overall in relation to the experiment, and using OpenWeather could actually reduce the viability since if I already have 3 preset alternatives, differences in real temperature would lead to differences in cost and comfort across runs. Thus, i added a preset outdoor temperature	
1/26/26	I tallied up my numbers and I forgot to account for the header row being a	

1 hour	row in every sheet. I was at least one scenario short in each scenario list, and in others I had simply not gotten close to the overall needed scenario count. I added placeholders up to the scenario counts I needed for each decision type and sheet, and started to fix the variation in my test sets. About 20% of my current scenarios in every decision type had 2 alternatives really close together (e.g. 71 and 72 degrees/2pm and 3pm/5 minutes and 6 minutes) and then one alternative that was widely off (e.g. 85 degrees, 4 am, 30 minutes) which doesn't really help test the AI; at this point, it is essentially only differentiating between 2 alternatives since one is useless and these 2 alternatives are essentially the same. Thus, I worked on spacing out the alternatives more evenly. Additionally, I ran my ground-truth calculator through an AI and found multiple edge cases where the output would end up being near 0 or 10 just due to extreme multiplier (for example, I was using an 60% penalty on complex scenarios for HVAC type)
2/1/26 2 hours	I realized that developing a complex type for appliance and shower in terms of daily decision making is practically impossible to score with a calculator, I also fully developed the <code>calculate_practicality_score</code> and <code>calculate_comfort_score</code> functions. I decided to focus on three major things; the practicality based on the extremity penalty, which is the mental willingness. This is not the same as the overall comfort of someone, as people are often willing to set temperatures where they are not comfortable based on other benefits. This penalty on extremity was calculated with continuous use likelihood. Comfort, on the other hand, focused on the overall number of people in the household and the actual distance the alternative was from ASHARE standards. Comfort
2/2/26 20 minutes	I tested the overall idea of complex scenarios within HVAC and decided to drop it; there is no good way to get a lot of variability and still have a set of rules where the scenario can be defined as complex.
2/3/26 5 minutes	Haven't actually worked on my laptop but I actually thought of doing an occupancy context for HVAC; how many hours the occupants are actually at home. Still allows for some further penalty while still being structured.
2/4/26 2 hours	I officially added <code>occupancy_context</code> to the HVAC scenarios. I removed preference temperature from shower scenarios, and added a difference between peak and off-peak rate for the appliance scenarios (it was previously just TOU rate). I also finished work on the the HVAC ground truth calculator, and tested it with the first 10 scenarios, which seem to have a decent overall spread and scores. I also finalized the prompt for pure-prompting.py. I also chose the

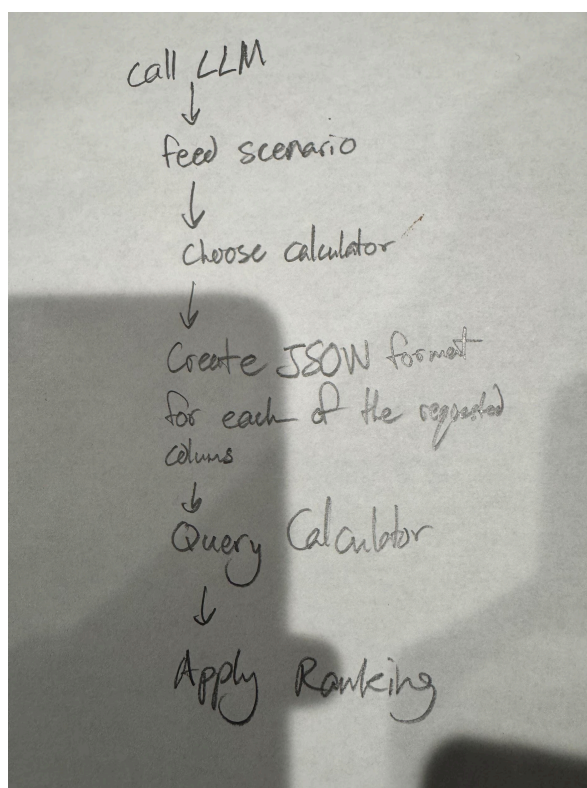
function types for each of the criterion (at least for HVAC, but I will likely use the same ones in the shower and appliance calculations as well). I chose linear for energy cost and environmental since most studies show that quantitative values are often perceived in a linear benefit fashion by users, and then logarithmic for comfort and practicality, since studies such as Ivanov et al. (2025) show that most behavioral economics with comfort use logarithmic curves. I have not chosen alpha values yet. My current (hopefully finalized) list of parameters needed for ground-truth for each decision types are:

(THIS WAS REFORMATTED INTO A TABLE LATER ON):

Parameters per Decision Type

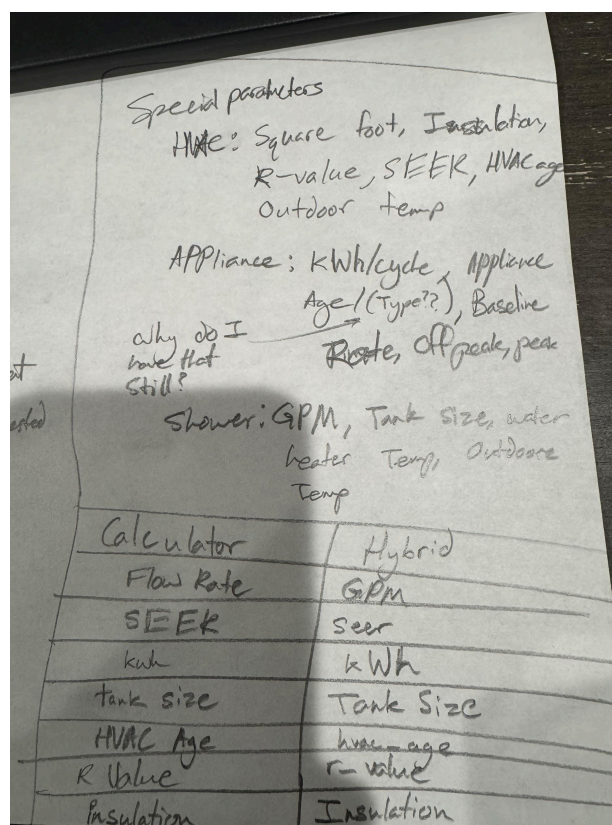
HVAC	APPLIANCE	SHOWER
Question	Description	Description
Location	Location	Location
Square Footage	Utility Budget	Occupants
Insulation	Appliance	Tank Size
Household Size	Housing Type	GPM
Utility Budget	Occupants	Utility Budget
Housing Type	Peak Rate	Housing Type
Outdoor Temp	Off-Peak Rate	Outdoor Temp
House Age	kwh/cycle	Water Heater Temp
Occupancy Context	Appliance Age/Type	Alternative 1
R-Value	Baseline Time	Alternative 2
HVAC Age	Alternative 1	Alternative 3
SEER	Alternative 2	
Alternative 1	Alternative 3	
Alternative 2		
Alternative 3		

2/5/26 2 hours	Today i worked entirely on the computemetrics.py file, which should now be able to compute kendalls tau, spearmans rho, MAE, RMSE, top-1, and top-2
2/7/26 3.5 hours	I finished my HVAC ground truth, which is ready to be tested. I then switched over to the RAG-enhanced architecture. I used the exact same functions I used in pure-prompting.py whenever possible, including the MAVT types. However, for RAG integration I chose to use chromaDB and the sentence-transformer model within it. I chose to retrieve 3 similar scenarios per call, and while I have the logic figured out for retrieval and formatting, I am still encountering multiple bugs with building the prompt since some of my parameter names are inconsistent. Placeholder prompts still present in some files, noted for resolution. Minor bug fix with a placeholder error.
2/8/26 1.5 hours	I was not able to import many functions from the other two architectures to the Hybrid architecture (literally only the query_openrouter function and the apply_mavt_ranking), but I was able to plan out the calculator integration. I decided that having 1 unified prompt for the AI (since it only has to extract parameters based on the question text) would be more efficient and is reasonable as it is not unfair to let hybrid “take advantage” of its efficiency, even if thatll mean lower token usage. I was also able to create the extraction function today (although not tested-extract_all_with_AI) which takes in the scenario and identifier, and then has the AI create a JSON format scenario for the ground-truth calculator to run on. However I have left placeholders for the parameter names as it is getting late and I have other work to finish



2/10/26
4 hours

today was unfortunately a bad day. First thing I did was fill in placeholders from yesterday. Then, I created `score_with_ground_truth`, which is all I worked on today. It is still not finished as there were a very very large amount of parameter name discrepancies (which is my fault, I should have used consistent naming) between the calculators and the csvs and the [hybrid.py](#) program. This took an extremely large amount of time. The below sheet was me listing what capitalization and name differences I had to fix the bugs:



I also started my slides for submission. I was able to finish every slide that did not deal with the actual results, which included hypothesis/engineering goals, background research, etc.

2/12/26
2 hours

Finished [hybrid.py](#) since the only thing left to do was code the main test set function. I also looked further into multiple sources by the DOE, EPA, EIA, ASHARE, as well as individual researchers to update many of my reference ranges. For appliance in particular I had not fact checked my ranges for environmental and energy criterion, so the estimated 5th and 95th percentiles were not accurate. I also went through each architecture and ground truth calculator to set the same emissions/kWh and base rate/kWh (0.6458 and 0.19 respectively for PA). I also had AI review my MAVT implementation, and I found no fundamental errors. Dr. Huang also noted a major but easily fixable error: I had been normalizing the values for each of the criterion before feeding it to MAVT, but the overall reason we had chosen MAVT as our method was because it can use bounds to normalize values to a criterion with a function. All I had to do was cut off some normalization code in each of the criterion calculation functions.

2/13/26
10 hours

The first thing I realized was that I had never actually finished the logic for calculating shower criterion scores. I spent a full hour finding reference ranges for this as well as finding studies on peoples preference

in terms of shower times. I then implemented each of the ranges, ran the groundtruth, and then noticed many of my values for the tested scenarios were off. This is one of the failed scenarios:

environme	energy cos	comfort	practicality
0	0	10	5
2.67	3.45	9.87	9.5
5	5	5	5
4.21	5.87	9.87	8.2

One of the alternatives had failed completely (all 5s) and the 1st alternative, which I had chosen since it represented the “average” decision (7 minutes, which according to my studies was supposed to be the usual range) was failing and throwing extreme values. Additionally, the comfort score for alternatives 2 should not have been as high as it was, as the alternative was for 2 minutes of showering. I thus had to go back and adjust all of my ranges. I realized that the problem was because I was using the flow rate as a factor for all 4 on accident, which was creating values that shouldn’t have been possible for kWh/lbs CO2 and was then getting normalized HEAVILY by an extremely large alpha value. The only reason that there were any values besides 0 and 10 was actually due to this large alpha value. I thus had to go back and fix many of the calculations. I also did similar testing for the other two calculators and found similar (but less severe) problems.

Thankfully, I only encountered 1 actual code error when testing if the three architectures would run; it was an issue with me calling the model from openrouter, unfortunately it took me awhile to actually understand that I had just misspelled the name of the model.

Once I ran all three of the architectures fully, though, I found a variety of issues. For one, both the hybrid and RAG architectures were throwing 0s for all of the appliance scenarios, which meant that the parsing was obviously failing but for some reason it was catching an error (which would have made every value a 5). I went back and checked the testscenarios file, and then realized that I had placed my appliance age in the house age column. I also started reviewing the other files with for similar discrepancies along with claude opus 4.6 and unfortunately found a very large amount:

- RAG and Testing scenario overlap (1 in HVAC, 3 in appliance, 10 in shower)
- Only 97 total testing scenarios
- RAG database scenarios missing columns that were supposed to be added (GT columns, since I had also today decided that the RAG database should get a full list of parameters to help the AI

contextualize the Ground truth values it was getting)

- GT datasets also missing columns (only 143 total)

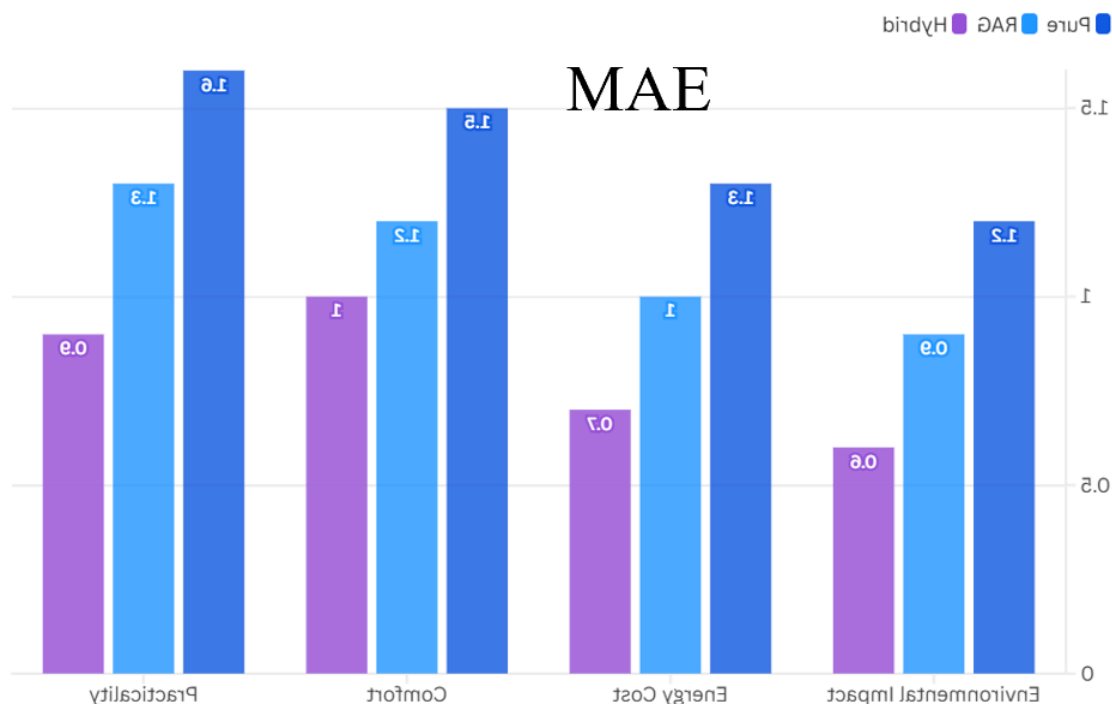
I then went through every sheet and added more scenarios where needed, created multiple extra sheets to see where the overlap was, repeatedly messed up the overlap, but finally finished sorting everything about 2 hours, along with the help of AI.

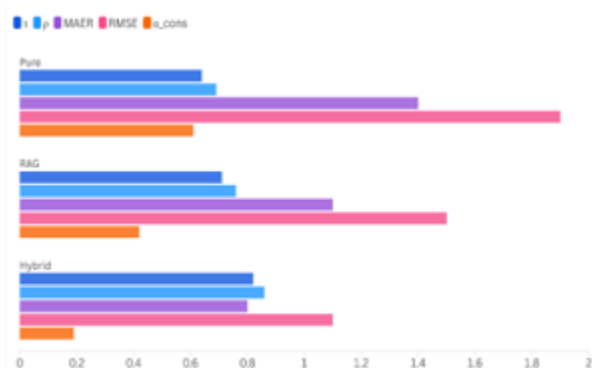
Once all of that was done, I was able to run every ground truth. Each of them were good besides the dryer scenarios for the appliance types, which were failing due to a parameter mismatch. Once i finally finished that, I was able to run all three ground truths to come up with a full ground truth list.

Then, I went back to the architectures. The RAG architecture was still having dependency problems, so I had to switch pythons to version 3.11.

Once I did that redownloaded all dependencies for each of the architectures, I was finally able to run all thre architectures successfully.

Once finished, I run [computemetrics.py](#). However, since I had been using multiple different files and filenames had been changed, I had to go back through and readjust all of the file names. I ended up with these metrics:





Overall Physical Metrics

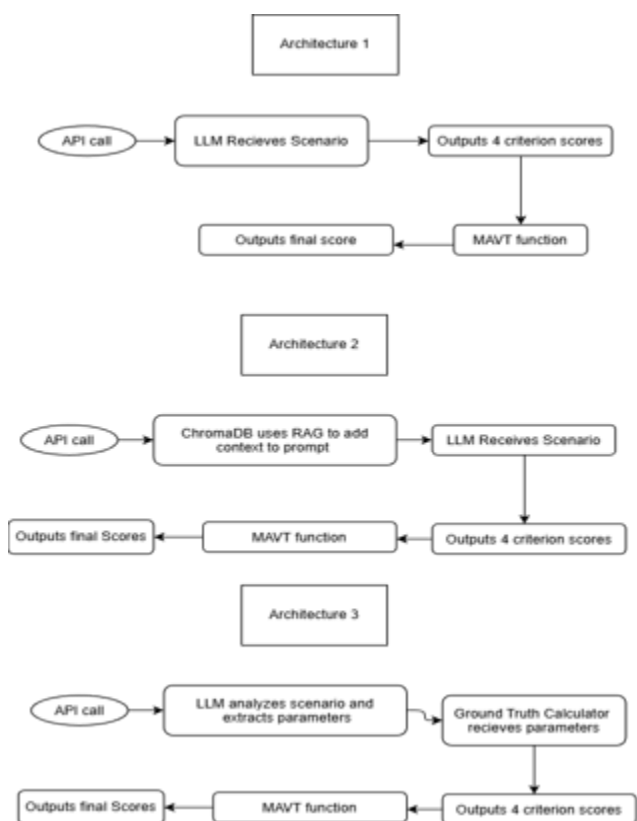
	Input Tokens (thousands)	Output Tokens (thousands)	Total Tokens (thousands)	API Calls	Total Time (seconds)	Success Rate (%)
Pure	113.2	10.2	123.4	297	324	100
RAG	241.5	12.2	253.7	297	367	100
Hybrid	80.8	15.9	96.7	99	259	58.6

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Architecture Tradeoffs (WITHOUT FAILED SCENARIOS)

Dimension	Pure	RAG	Hybrid
Accuracy	Low ($\tau=0.64$)	Medium ($\tau=0.71$)	High ($\tau=0.82$)
Consistency	Poor ($\sigma=0.61$)	Moderate ($\sigma=0.42$)	Excellent ($\sigma=0.19$)

I also finished my AI use section, which I updated to help with coding files since I had used it multiple times today in a time crunch. I also have created a architecture diagram I intend to update later:



A photo of my workspace:



2/14/26 5 minutes	Committed all files
2/19/26 1.5 hours	read EPA eGRID documentation more carefully and did one final overview of my values and citations for every number I used in the ground truth files. Found no major errors, other than my periods per month in the ground truth calculator which I initially had at 30 upped to 90 (8hours per day * 3 = 24, not just 1 per day). I also reorganized my files within files explorer since not everything was within the overall folder; <i>LLM-MCDA</i>
2/23/26 1 hour	Added baseline time to Appliance scenarios, deleted all extra files. I also realized that the questions for the scenarios were currently really basic (same question over and over with no hint of occupancy context or anything, so I spent most of today going through every file and rephrasing questions for all 3 decision types.
2/28/26 5 hours	Today I noticed a fundamental error in my architectures; for all three, I had been relying on the ground truth file sheets, which is to say I had been giving them all of the parameters with all of the specificity that the GT got. I updated each architecture to receive the more general information. I reran every architecture on the updated columns and new question types, and got very different scores. I also noticed while the files were running that the RAG shower scenarios and some of the Appliance scenarios were still throwing 0s, which actually ended up being due to a request threshold error. I changed the way ChromaDB received the scenarios (now only feeding columns rather than extra context as well) and was able to fix this error. I then reran the RAG architecture completely. When running calculatemetrics.py, however, I hit multiple errors. Essentially what I found on the 13th is that I had in a way hardcoded the scenario parsing by column number, and since I had changed the input for each of the architectures, they now had many less columns. I had also in the past implemented a hardcoded fix which combined some of my files (since I had a unorganized list on the 13th), which was also retrieving old files and wrong row numbers. I then spent about an hour fixing the calculatemetrics.py file and made it a lot less hardcoded, instead relying on the built-in decision type identifiers instead of exact column numbers. Throughout the entirety of this session for each file, I was also often implementing fixes for the time of day parsing in appliance types, since they were formatted differently for the architectures for better

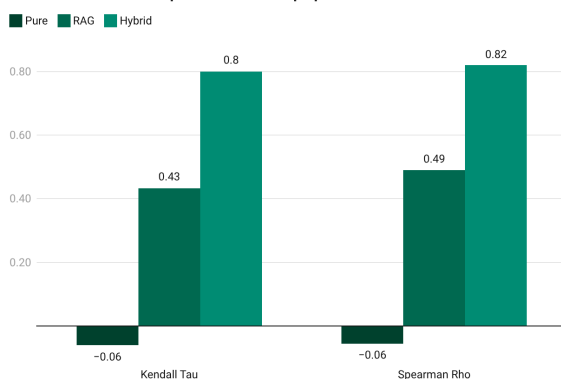
interpretability.

Unfortunately I was not able to run [calculatemetrics.py](#) today

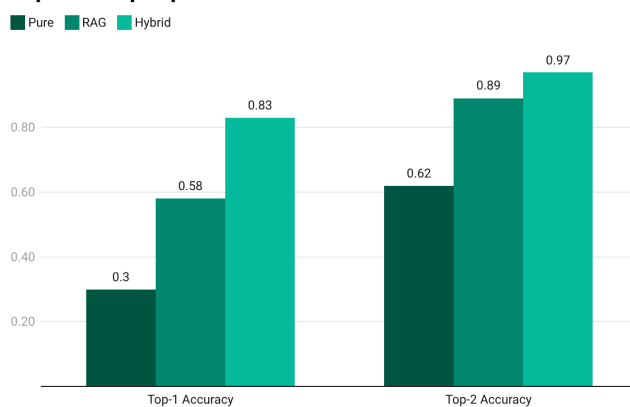
3/1/26
2 hours

I completed the last parsing issue with appliance times and was able to run [calculatemetrics.py](#) and got a finalized set of metrics. I then spent the rest of today converting these metrics in to more interpretable sets (I finished 3 today) . I also ended up with a 0% fail rate for each architecture, so I dropped it as a comparison and just will include that as a note. ***ALL LATER WORK WAS ONLY ON NOTEBOOK AND POSTER***

Kendall's τ & Spearman's ρ per Architecture



Top-1 vs Top-2 per Architecture



Overall MAE & RMSE per Architecture

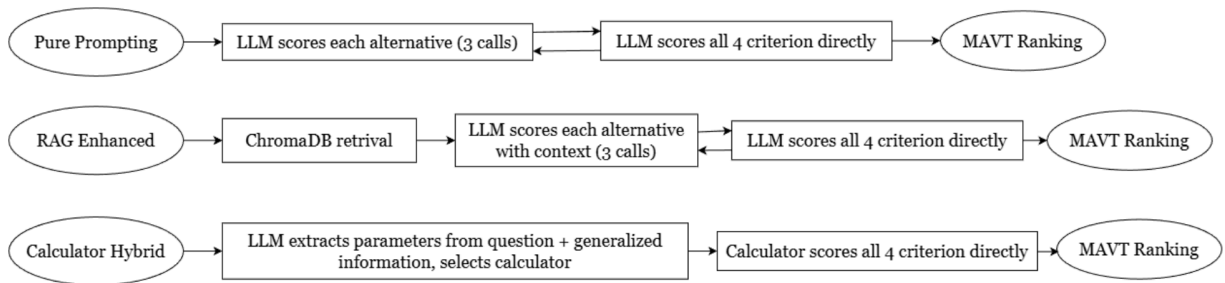


Analysis

***Values are rounded for graphs and analysis. Exact values for all metrics can be found in metrics.**

The Hybrid Architecture outscored both other architectures. The Pure Prompting architecture had almost random overall ranking and extreme error on every criterion across decision types. The RAG Enhanced architecture performed much better, but still not produce reliable levels of overall ranking

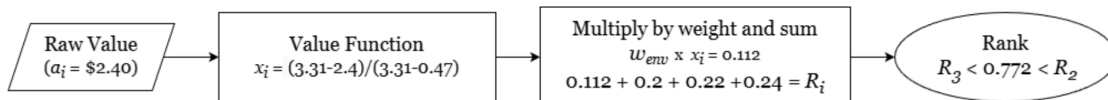
How the Architectures Work



MAVT Ranking

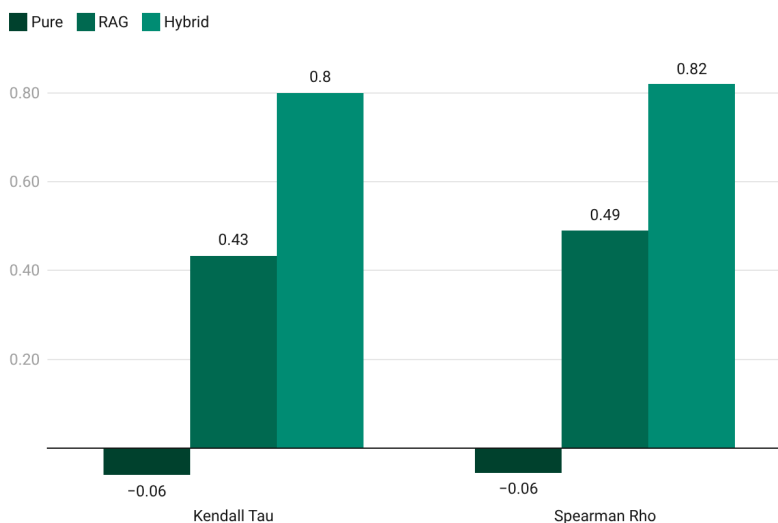
$$1. a_i \quad 2. F(a_i) = x_i \quad 3. \sum_{i=1}^n w_i x_i = R_i \quad 4. R_1 < R_3 < R_2$$

Example Environmental MAVT application



Kendall's τ & Spearman's ρ per Architecture

Measures ranking accuracy. Kendall's τ specifically counting how many individual pairs of alternatives are swapped in the wrong order, while Spearman's ρ looks at how far off each alternative's predicted rank position is from its true position overall, both on a scale from -1 (completely reversed) to 1 (perfect match).

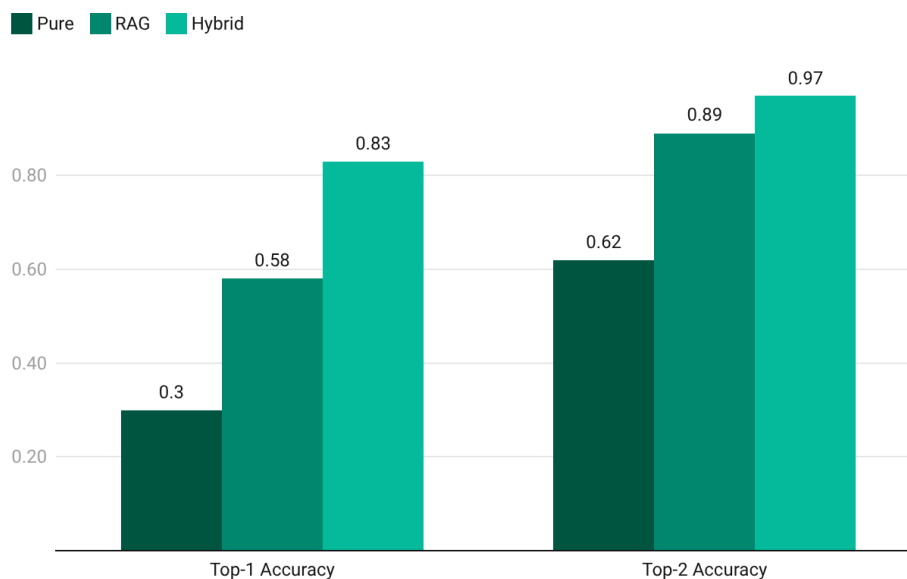


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The Hybrid architecture had a Kendall's τ of 0.80 and Spearman's ρ of 0.82. This shows that the hybrid architecture had both a correct overall ranking order, and when the ranking was incorrect, the incorrect placement of the alternative was not far off of the true best position. RAG Enhanced was moderate at $\tau = 0.43$, $\rho = 0.49$. Pure Prompting scored $\tau = -0.06$, $\rho = -0.055$, which means that it was statistically indistinguishable from random and it provided no reliable ranking information.

Top-1 vs Top-2 per Architecture

How often each architecture identifies the single best alternative/best two alternatives. End user would only see these decisions, thus in implementation these are the most important metrics.



Created with Datawrapper

Hybrid correctly identified the best alternative 83% of the time and identified the best alternative in its top 2 picks 97% of the time. RAG-Enhanced had 58% and 89% respectively. Pure Prompting reached only 30% Top-1 accuracy, which is below the average (33%) from picking randomly. While its Top-2 was better at 62%, the overall usefulness of the architecture is low.

Overall MAE & RMSE per Architecture

Compares two measures of average scoring error per architecture. MAE measures typical error size, while RMSE is more sensitive to occasional large mistakes.



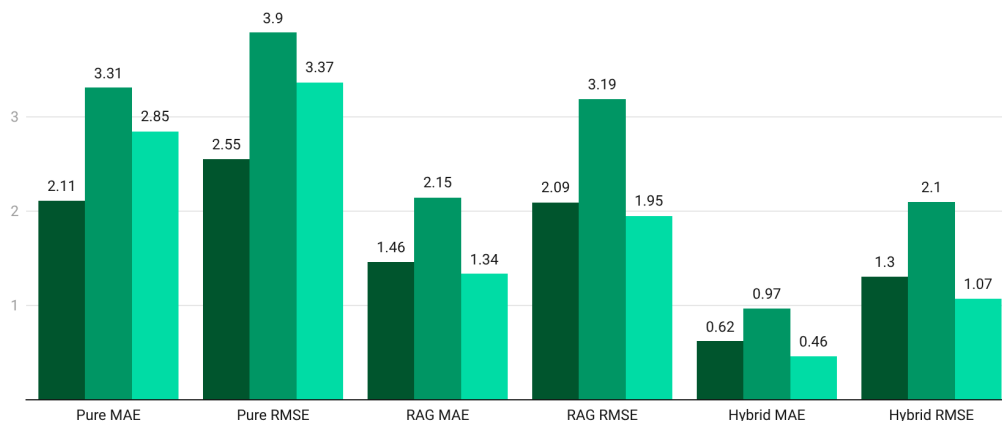
Created with Datawrapper

Hybrid produced the lowest overall MAE at 0.69, and RMSE at 1.52 (all on a 0 to 10 scale). While this metric isn't as important for the end user, it does help identify how far off the architecture was on average (all three had better than random error) and how many large errors it made. RAG-Enhanced was intermediate at MAE = 1.61, RMSE = 2.40. Pure Prompting had the highest error at MAE = 2.52, RMSE = 3.07. The gap between Hybrid and Pure Prompting represents roughly a 1.8-point average error reduction per scenario. While Pure Prompting had the largest overall error amount, it had the smallest gap between RMSE (large errors penalized more) and MAE.

MAE & RMSE by Decision Type per Architecture

Shows patterns in error amounts (average and large errors) across decision types, common among architectures

■ HVAC ■ Appliance ■ Shower

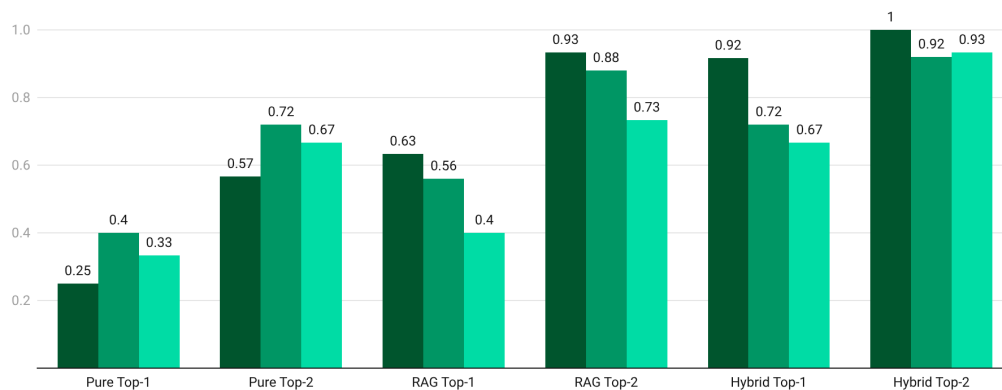


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Top-1 & Top-2 by Decision Type per Architecture

Shows whether architectures that miss the optimal alternative at least rank it second, broken down by decision type

■ HVAC ■ Appliance ■ Shower

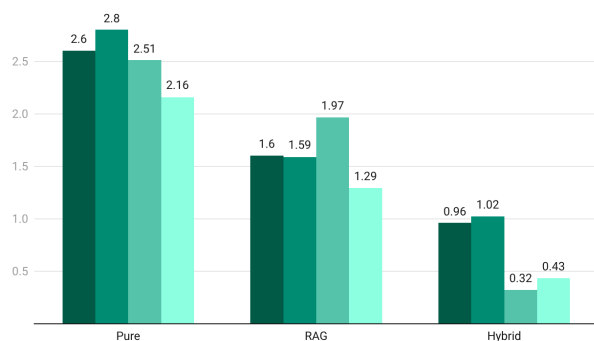


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MAE by Criterion per Architecture

Which of the four criterion each architecture scores most and least accurately on average across all 100 scenarios.

■ Energy Cost ■ Environmental ■ Comfort ■ Practicality

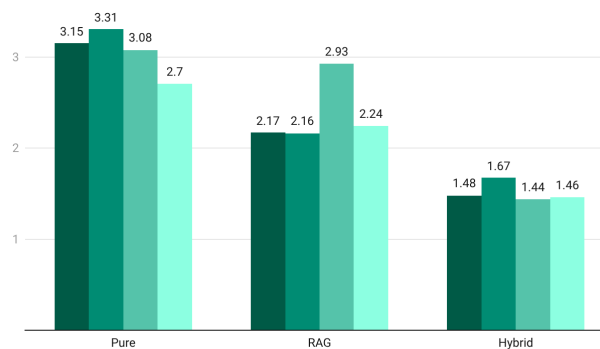


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RMSE by Criterion per Architecture

Shows Impact of large individual scoring errors per criterion, since RMSE penalizes outliers more heavily than MAE does.

■ Energy Cost ■ Environmental ■ Comfort ■ Practicality

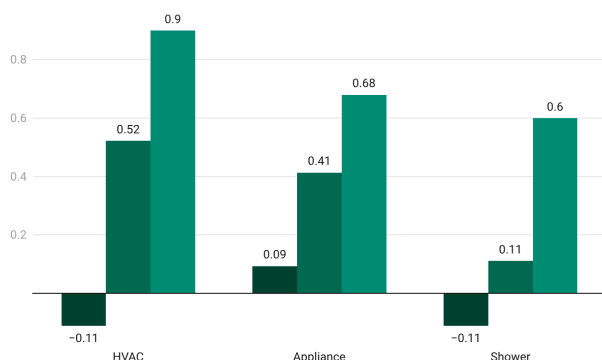


Created with Datawrapper

Kendall's τ by Decision Type per Architecture

Shows how different ranking order is from ground truth (scale -1 to 1) on each decision type

■ Pure ■ RAG ■ Hybrid

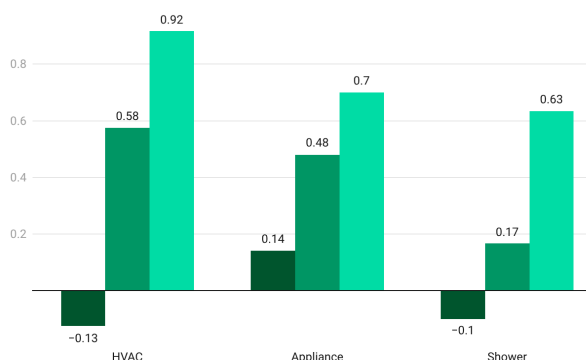


Created with Datawrapper

Spearman's ρ by Decision Type:

Shows how far off the alternatives were ranked relative to their best position (-1 to 1) on each decision type

■ Pure ■ RAG ■ Hybrid



Created with Datawrapper

The above 6 graphs show some very interesting patterns on the overall scoring biases of each architecture.

In HVAC scenarios (Top-1 = 92%, Top-2 = 100%, RMSE = 1.3, MAE = 0.62), where the deterministic calculator fully covered all pertinent parameters and additional estimation required very little LLM creativity, Hybrid performed best. In fact, on the comfort and practicality criterion, it had an overall MAE of 0.0! In the Appliance (Top-1 = 72%, Top-2 = 92%, RMSE = 2.1, MAE = 0.97) and Shower (Top-1 = 67%, Top-2 = 93%, RMSE = 1.07, MAE = 0.46) scenarios, where subjective factors like comfort and practicality necessitated partial AI estimation, it performed worse. RAG-Enhanced showed the same pattern among all of its Top-1, Top-2, RMSE, and MAE values as hybrid; its strongest ranking performance was also on HVAC even though Shower had lower RMSE and MAE, likely because HVAC scenarios have well-defined energy tradeoffs that map cleanly onto retrieved context.

Pure Prompting, on the other hand, performed the worst on HVAC of all decision types (MAE = 2.11, RMSE = 2.55, Top-1 = 25%, below the 33% random baseline;). Even though the errors were smaller in magnitude than shower and appliance scenarios, every criterion value showed a bias in the wrong direction. This pushed the overall ranking of the alternative in a different direction and thus had a much larger effect on Top-1, Top-2, Kendall's τ , and Spearman's ρ .

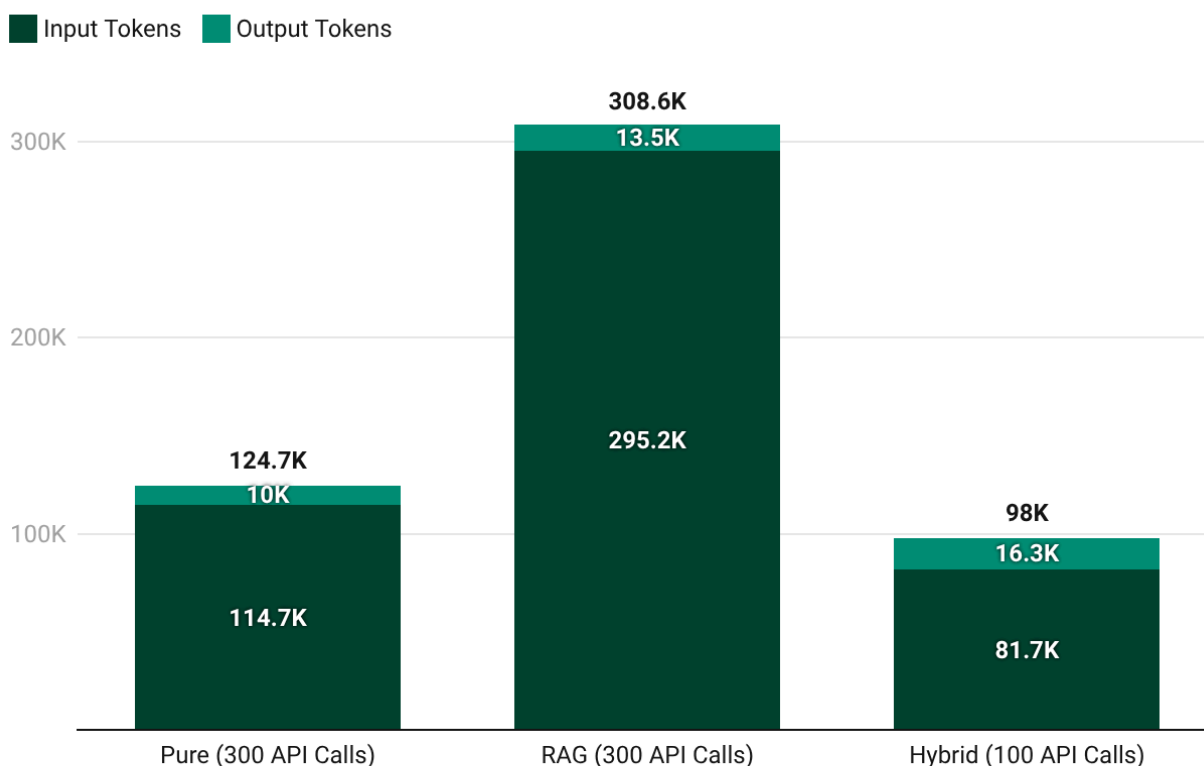
Appliance and Shower decision types are clearly the most important for exposing different failure points for each architecture. Pure Prompting scored highest on Appliance—its only decision type with a positive tau—suggesting LLM general knowledge about appliance tradeoffs is slightly more reliable than for

HVAC or Shower. However, its Appliance MAE (3.31) and RMSE (3.90) were the highest of any architecture-decision type combination. Despite having a lower MAE (1.34 vs. 1.46), RAG-Enhanced's Shower performance was noticeably worse than its HVAC performance. This suggests that the small Shower RAG database (5 scenarios) produced good average scores but failed to correctly rank alternatives within scenarios. Hybrid's Shower MAE (0.46) was the lowest of all decision types, but it showed the exact same ranking problem.

However, in all of these analyses, there are partial biases with the decision type itself and the parameters selected; for instance, HVAC scenarios are much simpler to extract parameters for than appliance scenarios, so the patterns between decision types are not solely attributable to the architecture's quality.

Token Usage per Architecture

Shows computing cost tradeoffs of each architecture by comparing input, output, and total token consumption.



Created with Datawrapper

While Hybrid only needed 100 API calls (one per scenario), Pure Prompting and RAG-Enhanced needed 300 calls across 100 scenarios (three per scenario, one per alternative). Because it processed all three options in a single prompt, Hybrid used the most output tokens per call despite making fewer calls (16.3K total output vs. 10.0K for Pure). Due to the retrieval context appended to each call, RAG-Enhanced's input token count (295.2K) was almost 2.6× that of Pure (114.7K).

Average Latency per Architecture

Shows efficiency by measuring mean response time per API call in seconds, independent of total run size.



Created with Datawrapper

Total Latency per Architecture

Cumulative processing time in seconds across all 100 scenarios. Represents larger scale use.



Created with Datawrapper

Hybrid's overall latency (275.6s across 100 scenarios) was lower than both Pure Prompting (437.1s) and RAG-Enhanced (480.8s) because each Hybrid call processed all three options in a single prompt rather than sequentially. However, due to the larger single prompt, Hybrid had the longest average latency (2,756 ms/call) when compared to Pure (1,457 ms/call) and RAG (1,603 ms/call). Out of the three, RAG-Enhanced had the highest overall latency. The retrieval step added an average of about 146 ms per call over Pure, which added up to a total overhead of 43.7 seconds over all 300 calls.

The latency and token results collectively point to a clear cost-accuracy tradeoff that favors Hybrid for any household decision-support tool meant for real-world deployment. While Pure and RAG's 300-call structure would require parallelization to achieve the same responsiveness at scale, Hybrid is fast enough for real-time interaction—a user describing their situation and receiving a ranked recommendation in less than 3 seconds is practically feasible. Because of its 2.6× token overhead over Pure without a corresponding accuracy gain ($\tau = 0.43$ vs. negative 0.06), RAG-Enhanced is the least efficient of the three; the retrieval cost is only justified if the database is large and well-matched enough to consistently return relevant context, which the shower results suggest is not guaranteed with small domain-specific databases. Practically speaking, the Hybrid's 0% failure rate in all 100 scenarios is also significant because a tool that occasionally fails to

generate a recommendation is worse than one that generates a slightly suboptimal one (and Hybrid fortunately does not suffer from this problem), and Pure and RAG's multi-call structure increases the surface area for partial failures even though none happened here.

Parameters by Decision Type for Architectures

Parameters given to Architectures (some generalized values)

HVAC	APPLIANCE	SHOWER
Question	Description	Description
Location	Location	Location
Square Footage	Utility Budget	Occupants
Insulation (Generalized)	Housing Type	Utility Budget
Occupants	Occupants	Housing Type
Utility Budget	Appliance Age (Range)	Outdoor Temp
Housing Type	Alternative 1	GPM (Generalized)
Outdoor Temp	Alternative 2	Alternative 1
House Age (Range)	Alternative 3	Alternative 2
Alternative 1		Alternative 3
Alternative 2		
Alternative 3		

Created with Datawrapper

Parameters per Decision Type for Ground Truth

Parameters given to each of the ground truth calculators (all specific values)

HVAC	APPLIANCE	SHOWER
Question	Description	Description
Location	Location	Location
Square Footage	Utility Budget	Occupants
Insulation	Appliance	Tank Size
Occupants	Housing Type	GPM
Utility Budget	Occupants	Utility Budget
Housing Type	Peak Rate	Housing Type
Outdoor Temp	Off-Peak Rate	Outdoor Temp
House Age	kwh/cycle	Water Heater Temp
Occupancy Context	Appliance Age	Alternative 1
R-Value	Baseline Time	Alternative 2
HVAC Age	Alternative 1	Alternative 3
SEER	Alternative 2	
Alternative 1	Alternative 3	
Alternative 2		
Alternative 3		

Created with Datawrapper

While architectures receive generalized or categorical equivalents of the same fields, such as insulation quality in place of R-value and appliance age range in place of exact age, ground truth calculators receive fully specified numerical

inputs. For instance, HVAC scenarios include exact R-value, HVAC age, and SEER rating, while appliance scenarios include precise kWh/cycle, peak and off-peak rates, and appliance age. The same pattern can be seen in shower scenarios, where architectures receive only a generalized GPM range while ground truth receives exact GPM and tank size. While all three of the architectures have to do the same level of interpretation, the Hybrid architecture is able to focus only on extraction and thus performs better. Additionally, Hybrid also doesn't have to deal with the main problem with most LLMs: specific numbers.

One conclusion is consistently supported by the data across all metrics: LLM scoring alone cannot take the place of physics-based calculation. With HVAC scenarios showed the best case scenarios, and Appliance and Shower representing its floor where partial AI estimation introduced residual ranking errors, Hybrid performed the best on all primary metrics ($\tau = 0.80$, Top-1 = 83%, Top-2 = 97%, MAE = 0.69). RAG-Enhanced showed that retrieval context significantly outperforms unassisted scoring ($\tau = 0.43$ vs. -0.06), but database size and quality continue to be a bottleneck, especially on Shower scenarios where its five-scenario database generated accurate average scores but failed within-scenario ranking. The most significant failure is revealed by Pure Prompting's nearly random overall performance ($\tau = -0.06$) and simultaneous lowest MAE on HVAC with worst Top-1 on HVAC: systematic directional bias in criterion scoring that compounds across alternatives and collapses ranking accuracy even when individual errors appear numerically modest. With fewer total tokens, lower total latency, and a directly deployable single-call structure, Hybrid is the only architecture among the three that is practical for real-world household decision support, according to the token and latency results.

Conclusion

The hypothesis was largely confirmed; Hybrid achieved the strongest performance across all accuracy metrics ($\tau=0.80$, $\rho=0.82$, Top-1=83%, MAE=0.69, RMSE=1.52), while Pure Prompting produced near-random rankings ($\tau=-0.06$, Top-1=25-40% depending on decision type) and RAG-Enhanced fell in between ($\tau=0.43$, Top-1=58%). That physics-based MCDA computation cannot be consistently replicated by end-to-end LLM scoring. Retrieval context is a useful tool, especially for HVAC scenarios where well-defined energy tradeoffs mapped cleanly onto the database (Top-1=63%, $\tau=0.52$), as demonstrated by RAG-Enhanced's improvement over Pure Prompting. Retrieval alone, however, was insufficient for criteria that required subjective judgment: Pure Prompting's Shower environmental MAE of 4.04 was the only criterion-level error in the dataset that was higher than RAG's Appliance comfort MAE of 3.88. The difference between Hybrid (MAE=0.97) and Pure Prompting (MAE=3.31) was the biggest of any decision type in appliance scenarios, where exact quantitative reasoning about TOU rates and kWh/cycle is required.

One prediction was false: Hybrid used only 98,016 total tokens compared to Pure's 124,709, defying the theory that Hybrid would be the second most token-intensive due to structured output requirements, even though Hybrid used more output tokens per call. Because retrieval context was added to each prompt, RAG used the most tokens overall (308,637), and the 100-call structure compared to 300 calls for Pure and RAG more than made up for it. The fact that Hybrid was simultaneously the fastest, most accurate, and most token-efficient architecture (275.6s total latency vs. 437.1s for Pure and 480.8s for RAG) makes it the most practical architecture for real-world household energy decision support. The only major barrier to using a Hybrid type architecture is the immense research and time needed to actually build the calculators used by the architecture; in cases where perfect decision choices are not always needed, a lower cost RAG model (Top-2 = 89%) could be useful.

Next Step

The main engineering focus currently is to improve Hybrid's performance for appliance and shower scenarios, which are significantly behind HVAC performance ($\tau=0.68$ and $\tau=0.60$ compared to $\tau=0.90$ for HVAC). The performance gap might be partly explained by the ability of occupancy context to enable completely deterministic extraction of comfort and practicality criteria, which resulted in a perfect 0.0 MAE for those criteria, while appliance and shower scenarios lack an analogous structural anchor. A concrete step forward would be to create scenario-type-specific extraction prompts rather than a general call for all types of scenarios. This would allow the appliance extractor to focus on scheduling context and the shower extractor to focus on duration and flow rates rather than trying to attend to all types simultaneously.

The other major generalizability constraint related to the parameters used is those specifically related to the state of Pennsylvania. The ground truth calculator uses a single data point for emissions factor (0.6458 lbs CO₂/kWh from EPA eGRID data) and PPL Electric's TOU pricing structure. As a result, architecture rankings might not generalize to other grid regions. A high renewables mix would compress environmental score differences between alternatives, which might affect which architecture performs best on this criterion. A larger set of parameters that includes all EPA eGRID subregions and real-time utility rates from the EIA API would allow us to test whether Hybrid's dominance is generalizable across energy contexts or is an artifact of the Pennsylvania energy mix.

The study was further limited to just one model. The Mistral Small 3.2 model was chosen for cost efficiency, but it is not clear if the advantages of Hybrid carry forward to more powerful models, nor is it clear if Pure Prompting can close the gap with a frontier model. If, in fact, a model with stronger reasoning capabilities can be expected to extract parameters and scoring criteria with high accuracy in one prompt, creating a physics based calculator for any decision type may be a waste of resources.

However, a more powerful model would be more costly. The same consideration can be applied to the converse situation, where if the parameters can be just as easily extracted by a lesser model, the advantages of Mistral Small 3.2, and its metrics, have not been optimized. The same 100 scenarios can be run against several model families to determine if the results represent architecture or model capabilities.

In the long term, the value of this tool lies in whether real households end up using this tool and whether it actually affects their decisions. The natural follow-up study on this topic would involve implementing a mobile interface for a group of households in Pennsylvania and tracking their decisions compared to the recommendations of this tool over a period of time.

AI USE

I used Claude, ChatGPT, and Perplexity throughout the duration of my project.

- ChatGPT was used during the early stages to brainstorm ideas and help discover gaps in research. However, I chose the final idea myself with a combination of AI and my father's recommendations.
- Second, Claude was used to generate initial code, but I tested and debugged all generated code.
- Third, Claude was used to review existing code implementations for logical errors and edge cases, including MAVT normalization and parameter extraction logic. I chose all final functions and logic.
- Fourth, Claude was used to reorganize and sort Excel scenario files I was dealing with scenario overlaps and incorrect scenario counts
- Fifth, since many papers used a lot of tiny variables to calculate load, Perplexity was used to help identify which parameters had a significant computational impact on ground truth scores. However, I cross checked the importance of every parameter with government sources.
- ChatGPT was used to write LaTeX script (formatting)
- Finally, Perplexity was used to identify candidate sources for the literature review; all sources were independently read and verified by the researcher before citation.

Examples for each type of prompt are provided below.

Example prompt used for scenario/file analysis:

Okay I had a few thoughts. first, I dont think we should preset values in the alternatives; since we rely on real time weather data, the effect of each temperature value would be different from time to time when testing. I instead think there should be a preset formula, such as "outdoor temp-[preset value for alternative]" in cooling or vice versa in heating. This can be done in the calculator, so all that really has to be passed in is the preset value. Also, There needs to be 100 scenarios, not 80. I want 60 simple 40 complex, complex just meaning that there is more than one variable being considered, such as your "medium" scenarios from earlier. Now. can you take the second file attached and distribute the scenarios I have created in it accordingly?. Make sure that there is 60%/40% split in this as well.

I have attached the Scenario list to be fed to the AIs (that one is partially done), as well as the scenario list for ground truth. These are both for HVAC. Can you make sure the scenarios are lined up and have the same alternatives, question text, etc.

Fundamental correction prompt example:

Well that's not really how MCDAs work. Just because one alternative is the best or worst in a certain criterion category does not mean it should automatically receive the best or worst score possible. This is because the other criterion categories matter too, which is why we rank each alternative across multiple criterion; these criterion have their own weightings, and based on how much each criterion matters (weighting) and the value assigned for each alternative for each criterion, we calculate the best alternative across all criterion.

Brainstorming example (very early in ideation):

I am trying to conduct an independent research project and write a paper on it about how optimized people are. What I want to do in essence is somehow quantify most people's live on a scale with a variety of factors. I realize I might have to narrow the topic down - any suggestions? I want to conduct studies of what people are doing, and then compare these actions to all of the worse things that people may be doing - for example, people usually go to school everyday, even if they don't want to. Instead, they could have decided to go to a bank and attempt to rob it, fail, and permanently mess up their lives. Obviously there are too many things to consider, so how do you think I could make this more plausible. i do want to conduct somewhat original evidence though. I realize that quantifying the metrics is hard, I need some help with that. I was thinking things like net overall long term and short term benefits, happiness, and quality of life - this could be like how going to the gym improves your health over time, but it can also make some people unhappy, so there would be points added and subtracted.

Coding example:

I am not going to include reference values in either prompt [COMMENT FOR NOTEBOOK: THIS IS FOR THE RAG ARCHITECTURE]. it defeats the purpose. I made it like this; in the CSVs, I will not include ANY unnecessary suffixes or prefixes.: Alternative formats must match exactly: - HVAC: "72", "76", "80" (No suffix) - Appliance: "7pm", "10pm", "2am" - Shower: "5", "10", "15" (No suffix)

Also, I know our current code does these parameters: "Appliance Age/Type": "<age> OR <type>". I do not believe this is the right way; this introduces large amounts of variance for the AI and the ground truth calculator. I would like to remove the current parameter list in the code (e.g. top_std, etc.). and we have to do that for every factor that we have multiple specialized types for, so tell me what else to change/add.

IM not including decision_type_hint, please help me remove that too. I have removed it from the code, but let me know if there were any usages of that parameter I did not account for

also does the HVAC and respective functions in architecture files account for parameter input for occupancy context? If not, we have to include that calculation for the HVAC calculator. The current options for occupancy context are "occupied_all_day"/"unoccupied_<HOURS UNOCCUPIED."/"occupied_sleep". Let me know if you see other formats; we have to remove those.

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Appendix

Requirments.txt:

annotated-doc==0.0.4
annotated-types==0.7.0
anyio==4.12.1
attrs==25.4.0
backoff==2.2.1
bcrypt==5.0.0
build==1.4.0
certifi==2026.2.25
charset-normalizer==3.4.4
chromadb==1.5.2
click==8.3.1
colorama==0.4.6
distro==1.9.0
durationpy==0.10
et_xmlfile==2.0.0
filelock==3.24.3
flatbuffers==25.12.19
fsspec==2026.2.0
googleapis-common-protos==1.72.0
grpcio==1.78.0
h11==0.16.0
hf-xet==1.3.2
httpcore==1.0.9
httptools==0.7.1
httpx==0.28.1
huggingface_hub==1.5.0
idna==3.11
importlib_metadata==8.7.1
importlib_resources==6.5.2
Jinja2==3.1.6
joblib==1.5.3
jsonschema==4.26.0
jsonschema-specifications==2025.9.1

kubernetes==35.0.0
markdown-it-py==4.0.0
MarkupSafe==3.0.3
mdurl==0.1.2
mmh3==5.2.0
mpmath==1.3.0
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oauthlib==3.3.1
onnxruntime==1.24.2
openpyxl==3.1.5
opentelemetry-api==1.39.1
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opentelemetry-exporter-otlp-proto-grpc==1.39.1
opentelemetry-proto==1.39.1
opentelemetry-sdk==1.39.1
opentelemetry-semantic-conventions==0.60b1
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overrides==7.7.0
packaging==26.0
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posthog==5.4.0
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pybase64==1.4.3
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pydantic_core==2.41.5
Pygments==2.19.2
PyPika==0.51.1
pyproject_hooks==1.2.0
python-dateutil==2.9.0.post0
python-dotenv==1.2.1
PyYAML==6.0.3
referencing==0.37.0
regex==2026.2.19
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requests-oauthlib==2.0.0
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rpds-py==0.30.0

safetensors==0.7.0
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scipy==1.17.1
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uvicorn==0.41.0
watchfiles==1.1.1
websocket-client==1.9.0
websockets==16.0
zipp==3.23.0

Github URL:

Contains all raw scenario lists, code, and other project files.

<https://github.com/AN00927/LLM-MCDA>

Ground truth and Architecture Input Parameters:

Parameters by Decision Type for Architectures

Parameters given to Architectures (some generalized values)

HVAC	APPLIANCE	SHOWER
Question	Description	Description
Location	Location	Location
Square Footage	Utility Budget	Occupants
Insulation (Generalized)	Housing Type	Utility Budget
Occupants	Occupants	Housing Type
Utility Budget	Appliance Age (Range)	Outdoor Temp
Housing Type	Alternative 1	GPM (Generalized)
Outdoor Temp	Alternative 2	Alternative 1
House Age (Range)	Alternative 3	Alternative 2
Alternative 1		Alternative 3
Alternative 2		
Alternative 3		

Created with Datawrapper

Parameters per Decision Type for Ground Truth

Parameters given to each of the ground truth calculators (all specific values)

HVAC	APPLIANCE	SHOWER
Question	Description	Description
Location	Location	Location
Square Footage	Utility Budget	Occupants
Insulation	Appliance	Tank Size
Occupants	Housing Type	GPM
Utility Budget	Occupants	Utility Budget
Housing Type	Peak Rate	Housing Type
Outdoor Temp	Off-Peak Rate	Outdoor Temp
House Age	kwh/cycle	Water Heater Temp
Occupancy Context	Appliance Age	Alternative 1
R-Value	Baseline Time	Alternative 2
HVAC Age	Alternative 1	Alternative 3
SEER	Alternative 2	
Alternative 1	Alternative 3	
Alternative 2		
Alternative 3		

Created with Datawrapper

MAVT Function Options:

Possible types of single-attribute value functions		
Type	Increasing value functions	Decreasing value functions
Linear	$u(x) = \frac{x - x_{min}}{x_{max} - x_{min}}$	$u(x) = \frac{x_{max} - x}{x_{max} - x_{min}}$
Polynomial	$u(x) = \left(\frac{x - x_{min}}{x_{max} - x_{min}} \right)^a$ For any $a > 0$: if $a > 1$, the function is convex downward (concave upward); if $0 < a < 1$, the function is convex upward (concave downward); a symmetric reflection occurs if $a \rightarrow \frac{1}{a}$.	$u(x) = \left(\frac{x_{max} - x}{x_{max} - x_{min}} \right)^a$
Exponential	$u(x) = \frac{1 - \exp\left(a \cdot \frac{x - x_{min}}{x_{max} - x_{min}}\right)}{1 - \exp(a)}$ For any $a \neq 0$: if $a > 0$, the function is convex downward (concave upward); if $a < 0$, the function is convex upward (concave downward); a symmetric reflection occurs if $a \rightarrow -a$.	$u(x) = \frac{1 - \exp\left(a \cdot \frac{x_{max} - x}{x_{max} - x_{min}}\right)}{1 - \exp(a)}$
Logarithmic	$u(x) = \frac{\ln\left(a \cdot \frac{x - x_{min}}{x_{max} - x_{min}} + 1\right)}{\ln(a+1)}$ For any $a > -1$: if $a > 0$, the function is convex upward (concave downward); if $-1 < a < 0$, the function is convex downward (concave upward); a symmetric reflection occurs if $a \rightarrow (1 - a)$.	$u(x) = \frac{\ln\left(a \cdot \frac{x_{max} - x}{x_{max} - x_{min}} + 1\right)}{\ln(a+1)}$

architecture	decision_type	metric	value
Pure	Overall	energy_cost_MAE	2.6037
Pure	Overall	energy_cost_RMSE	3.1508
Pure	Overall	environmental_MAE	2.8042
Pure	Overall	environmental_RMSE	3.3056
Pure	Overall	comfort_MAE	2.5146
Pure	Overall	comfort_RMSE	3.075
Pure	Overall	practicality_MAE	2.1619
Pure	Overall	practicality_RMSE	2.7043
Pure	Overall	overall_MAE	2.5211
Pure	Overall	overall_RMSE	3.0669
Pure	Overall	kendall_tau	-0.06
Pure	Overall	spearman_rho	-0.055
Pure	Overall	top1_accuracy	0.3
Pure	Overall	top2_accuracy	0.62
Pure	Overall	n_scenarios_evaluated	100
Pure	HVAC	energy_cost_MAE	2.2537
Pure	HVAC	energy_cost_RMSE	2.7489
Pure	HVAC	environmental_MAE	2.3763
Pure	HVAC	environmental_RMSE	2.8233
Pure	HVAC	comfort_MAE	2.1877
Pure	HVAC	comfort_RMSE	2.6161
Pure	HVAC	practicality_MAE	1.6232
Pure	HVAC	practicality_RMSE	1.9135
Pure	HVAC	overall_MAE	2.1102
Pure	HVAC	overall_RMSE	2.5511
Pure	HVAC	kendall_tau	-0.1111
Pure	HVAC	spearman_rho	-0.125
Pure	HVAC	top1_accuracy	0.25
Pure	HVAC	top2_accuracy	0.5667
Pure	HVAC	n_scenarios_evaluated	60
Pure	Appliance	energy_cost_MAE	3.0725
Pure	Appliance	energy_cost_RMSE	3.6609

Pure	Appliance	environmental_MAE	3.0908
Pure	Appliance	environmental_RMSE	3.6515
Pure	Appliance	comfort_MAE	3.5869
Pure	Appliance	comfort_RMSE	4.1978
Pure	Appliance	practicality_MAE	3.4985
Pure	Appliance	practicality_RMSE	4.0531
Pure	Appliance	overall_MAE	3.3122
Pure	Appliance	overall_RMSE	3.8982
Pure	Appliance	kendall_tau	0.0933
Pure	Appliance	spearman_rho	0.14
Pure	Appliance	top1_accuracy	0.4
Pure	Appliance	top2_accuracy	0.72
Pure	Appliance	n_scenarios_evaluated	25
Pure	Shower	energy_cost_MAE	3.2227
Pure	Shower	energy_cost_RMSE	3.6908
Pure	Shower	environmental_MAE	4.0384
Pure	Shower	environmental_RMSE	4.329
Pure	Shower	comfort_MAE	2.0349
Pure	Shower	comfort_RMSE	2.5087
Pure	Shower	practicality_MAE	2.0891
Pure	Shower	practicality_RMSE	2.5939
Pure	Shower	overall_MAE	2.8463
Pure	Shower	overall_RMSE	3.3684
Pure	Shower	kendall_tau	-0.1111
Pure	Shower	spearman_rho	-0.1
Pure	Shower	top1_accuracy	0.3333
Pure	Shower	top2_accuracy	0.6667
Pure	Shower	n_scenarios_evaluated	15
RAG	Overall	energy_cost_MAE	1.6043
RAG	Overall	energy_cost_RMSE	2.1698
RAG	Overall	environmental_MAE	1.5912
RAG	Overall	environmental_RMSE	2.1622
RAG	Overall	comfort_MAE	1.9696

RAG	Overall	overall_RMSE	2.3962
RAG	Overall	kendall_tau	0.4333
RAG	Overall	spearman_rho	0.49
RAG	Overall	top1_accuracy	0.58
RAG	Overall	top2_accuracy	0.89
RAG	Overall	n_scenarios_evaluated	100
RAG	HVAC	energy_cost_MAE	2.036
RAG	HVAC	energy_cost_RMSE	2.5325
RAG	HVAC	environmental_MAE	2.0366
RAG	HVAC	environmental_RMSE	2.5332
RAG	HVAC	comfort_MAE	1.4068
RAG	HVAC	comfort_RMSE	2.0549
RAG	HVAC	practicality_MAE	0.3719
RAG	HVAC	practicality_RMSE	0.6899
RAG	HVAC	overall_MAE	1.4628
RAG	HVAC	overall_RMSE	2.0934
RAG	HVAC	kendall_tau	0.5222
RAG	HVAC	spearman_rho	0.575
RAG	HVAC	top1_accuracy	0.6333
RAG	HVAC	top2_accuracy	0.9333
RAG	HVAC	n_scenarios_evaluated	60
RAG	Appliance	energy_cost_MAE	0.7987
RAG	Appliance	energy_cost_RMSE	1.2788
RAG	Appliance	environmental_MAE	0.9119
RAG	Appliance	environmental_RMSE	1.3995
RAG	Appliance	comfort_MAE	3.884
RAG	Appliance	comfort_RMSE	4.78
RAG	Appliance	practicality_MAE	2.994
RAG	Appliance	practicality_RMSE	3.7706
RAG	Appliance	overall_MAE	2.1471
RAG	Appliance	overall_RMSE	3.1883
RAG	Appliance	kendall_tau	0.4133

RAG	Appliance	top1_accuracy	0.56
RAG	Appliance	top2_accuracy	0.88
RAG	Appliance	n_scenarios_evaluated	25
RAG	Shower	energy_cost_MAE	1.22
RAG	Shower	energy_cost_RMSE	1.734
RAG	Shower	environmental_MAE	0.9416
RAG	Shower	environmental_RMSE	1.4947
RAG	Shower	comfort_MAE	1.03
RAG	Shower	comfort_RMSE	1.4571
RAG	Shower	practicality_MAE	2.1522
RAG	Shower	practicality_RMSE	2.8054
RAG	Shower	overall_MAE	1.3359
RAG	Shower	overall_RMSE	1.9516
RAG	Shower	kendall_tau	0.1111
RAG	Shower	spearman_rho	0.1667
RAG	Shower	top1_accuracy	0.4
RAG	Shower	top2_accuracy	0.7333
RAG	Shower	n_scenarios_evaluated	15
Hybrid	Overall	extraction_failure_rate	0
Hybrid	Overall	calculation_failure_rate	0
Hybrid	Overall	total_failure_rate	0
Hybrid	Overall	n_extraction_failures	0
Hybrid	Overall	n_calculation_failures	0
Hybrid	Overall	n_total_arch_scenarios	100
Hybrid	Overall	energy_cost_MAE	0.9622
Hybrid	Overall	energy_cost_RMSE	1.4777
Hybrid	Overall	environmental_MAE	1.0238
Hybrid	Overall	environmental_RMSE	1.6745
Hybrid	Overall	comfort_MAE	0.3226
Hybrid	Overall	comfort_RMSE	1.4366
Hybrid	Overall	practicality_MAE	0.4332
Hybrid	Overall	practicality_RMSE	1.4592

Hybrid	Overall	overall_RMSE	1.515				
Hybrid	Overall	kendall_tau	0.8				
Hybrid	Overall	spearman_rho	0.82				
Hybrid	Overall	top1_accuracy	0.83				
Hybrid	Overall	top2_accuracy	0.97				
Hybrid	Overall	n_scenarios_evaluated	100				
Hybrid	HVAC	energy_cost_MAE	1.1858				
Hybrid	HVAC	energy_cost_RMSE	1.7268	Hybrid	Appliance	top1_accuracy	0.72
Hybrid	HVAC	environmental_MAE	1.3083	Hybrid	Appliance	top2_accuracy	0.92
Hybrid	HVAC	environmental_RMSE	1.9557	Hybrid	Appliance	n_scenarios_evaluated	25
Hybrid	HVAC	comfort_MAE	0	Hybrid	Shower	energy_cost_MAE	0.4393
Hybrid	HVAC	comfort_RMSE	0	Hybrid	Shower	energy_cost_RMSE	0.7005
Hybrid	HVAC	practicality_MAE	0	Hybrid	Shower	environmental_MAE	0.2711
Hybrid	HVAC	practicality_RMSE	0	Hybrid	Shower	environmental_RMSE	0.52
Hybrid	HVAC	overall_MAE	0.6235	Hybrid	Shower	comfort_MAE	0.0573
Hybrid	HVAC	overall_RMSE	1.3045	Hybrid	Shower	comfort_RMSE	0.2259
Hybrid	HVAC	kendall_tau	0.9	Hybrid	Shower	practicality_MAE	1.0693
Hybrid	HVAC	spearman_rho	0.9167	Hybrid	Shower	practicality_RMSE	1.943
Hybrid	HVAC	top1_accuracy	0.9167	Hybrid	Shower	overall_MAE	0.4593
Hybrid	HVAC	top2_accuracy	1	Hybrid	Shower	overall_RMSE	1.0709
Hybrid	HVAC	n_scenarios_evaluated	60	Hybrid	Shower	kendall_tau	0.6
Hybrid	Appliance	energy_cost_MAE	0.7395	Hybrid	Shower	spearman_rho	0.6333
Hybrid	Appliance	energy_cost_RMSE	1.1331	Hybrid	Shower	top1_accuracy	0.6667
Hybrid	Appliance	environmental_MAE	0.7924	Hybrid	Shower	top2_accuracy	0.9333
Hybrid	Appliance	environmental_RMSE	1.3686	Hybrid	Shower	n_scenarios_evaluated	15
Hybrid	Appliance	comfort_MAE	1.2559				
Hybrid	Appliance	comfort_RMSE	2.8679				
Hybrid	Appliance	practicality_MAE	1.0913				
Hybrid	Appliance	practicality_RMSE	2.5004				
Hybrid	Appliance	overall_MAE	0.9698				
Hybrid	Appliance	overall_RMSE	2.0996				
Hybrid	Appliance	kendall_tau	0.68				
Hybrid	Appliance	spearman_rho	0.7				

Sample HVAC AI Scenario:

Question: The house will be empty for 8 hours today. What AC temperature should I set?

Location: Chester Springs, PA

Square Footage: 1800

Insulation: Poor

Household Size: 3

Utility Budget: \$200

Housing Type: Condo

Outdoor Temperature: 88°F

House Age: 20+ years

Alternative 1: 78°F

Alternative 2: 80°F

Alternative 3: 83°F

Corresponding HVAC Calculator Scenario:

Question: The house will be empty for 8 hours today. What AC temperature should I set?

Location: Chester Springs, PA

Square Footage: 1800

Insulation: Poor

Household Size: 3

Utility Budget: \$200

Housing Type: Condo

Outdoor Temperature: 88°F

House Age: 22 years

Occupancy Context: unoccupied_8hr

R-Value: 11

HVAC Age: 18 years

SEER: 12

Alternative 1: 78°F

Alternative 2: 80°F

Alternative 3: 83°F

Sample Appliance AI Scenario:

Question: Dinner's done around 5 PM. What's the best time to run the dishwasher?

Location: Allentown, PA

Utility Budget: \$150

Appliance Age Range: 1 to 3 years

Housing Type: Single-family

Occupants: 2

Alternative 1: Run dishwasher at 5:00 PM

Alternative 2: Run dishwasher at 7:00 PM

Alternative 3: Run dishwasher at 4:00 PM

Corresponding Appliance Calculator Scenario:

Question: Dinner's done around 5 PM. What's the best time to run the dishwasher?

Location: Allentown, PA

Utility Budget: \$150

Appliance: Dishwasher

Housing Type: Single-family

Occupants: 2

Peak Rate: \$0.19

Off-Peak Rate: \$0.08

Energy Use per Cycle: 1.25 kWh

Appliance Age/Type: 2

Baseline Time: 5:00 PM

Alternative 1: 5:00 PM

Alternative 2: 7:00 PM

Alternative 3: 4:00 PM

Sample Shower AI Scenario:

Question: How long should my showers be?

Location: Pittsburgh, PA

Occupants: 2

Flow Rate: Standard

Utility Budget: \$200

Housing Type: Condo

Outdoor Temperature: 35°F

Alternative 1: 5 minutes

Alternative 2: 8 minutes

Alternative 3: 12 minutes

Corresponding Shower Calculator Scenario:

Question: How long should my showers be?

Location: Pittsburgh, PA

Occupants: 2

Water Heater: Electric

Tank Size: 40 gallons

Flow Rate: 2.5 GPM

Utility Budget: \$200

Housing Type: Apartment

Outdoor Temperature: 35°F

Water Heater Temperature: 120°F

Alternative 1: 5 minutes

Alternative 2: 8 minutes

Alternative 3: 12 minutes

Dr. River Huang Correspondence:

Project on MCDAs for Household Optimization

Inbox x

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A

Ahaan Nigam
Dear Dr. Huang, My name is Ahaan Nigam, and I am a high school student at Downingtown East High School from Chester Springs, PA. I am working on a research proj

Wed, Dec 10, 2025, 5:38 PM ☆

H

Huang, He <river.huang@psi.ch>
to me ▾

Thu, Dec 11, 2025, 6:10 AM ★ 😊 ↶ ⋮

Dear Ahaan,

It is very nice to hear from you.

An AI-assisted MCDA tool is also something I have been thinking about, and I was pleased to hear that you are developing such a tool. I agree that it could be very helpful. At the same time, the design of a tool that integrates LLMs is quite delicate, as the AI can also influence users' behavior through the way it presents its outputs and recommendations. I have read your provided documents, and I'm happy to know more details of your plan.

I would be happy to discuss this topic with you in more depth. If you agree, we could schedule an online meeting in the coming weeks. Please let me know your thoughts.

Best regards,
River

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Schweiz

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Reference Ranges Table:

Decision Type	Criterion	Min	Max	Decreasing
HVAC	Energy Cost	0.47	3.31	Yes
HVAC	Environmental	1.6	11.25	Yes
HVAC	Comfort	0	10	No
HVAC	Practicality	1.5	10	No
Appliance	Energy Cost	0.02	0.9	Yes
Appliance	Environmental	0.09	3.83	Yes
Appliance	Comfort	0	10	No
Appliance	Practicality	1.5	10	No
Shower	Energy Cost	0.2	1.4	Yes
Shower	Environmental	1.1	5.9	Yes
Shower	Comfort	0	10	No
Shower	Practicality	1.5	10	No

Value Function	Notes
Linear	5th-95th percentile from actual dataset distribution
Linear	Derived from energy bounds × PA emissions factor (0.6458 lbs/kWh)
Logarithmic a=1.5	
Logarithmic a=1.2	Floor of 1.5
Linear	HE washer off-peak min; standard electric dryer at peak max
Linear	Adjusted to dataset; both bounds shifted from theoretical
Logarithmic a=1.5	
Logarithmic a=1.2	Floor from calculation
Linear	Short summer efficient min; long winter high-temp max
Linear	Derived from energy bounds × PA emissions factor (0.6458 lbs/kWh)
Logarithmic a=1.5	
Logarithmic a=1.2	Floor prevents zero practicality

